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Inventor(s)	BERMAN; Michael et al.

COMPUTERIZED TOOL PATH GENERATION

Abstract

An automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate a machined object from a workpiece, the machined object being configured to facilitate subsequent finishing into a finished object, the method including defining a surface of the finished object, defining an offset surface defining an inner limiting surface of the machined object, defining a scallop surface defining an outer limiting surface of the machined object and calculating a tool path for the milling machine which produces multiple step-up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object all lie between the inner limiting surface and the outer limiting surface and the number of step-up cuts in the workpiece and the areas cut in each of the step-up cuts are selected to generally minimize the amount of workpiece material that is removed from the workpiece.

Inventors: **BERMAN; Michael (Kfar Yona, IL), OSOVLANSKI; Doron (Givat Shmuel, IL), CALDERONE; Christopher Matthew (Levittown, PA), CALDERONE; Anthony Joseph (Yardley, PA)**

Applicant: **SOLIDCAM LTD.** (Or Yehuda, IL)

Family ID: **49775079**

Assignee: **SOLIDCAM LTD. (Or Yehuda, IL)**

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Background/Summary

REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of U.S. patent application Ser. No. 13/916,918, filed Jun. 13, 2013, which is a divisional of U.S. patent application Ser. No. 13/036,726, filed Feb. 28, 2011, published on Aug. 30, 2012 as U.S. Published Patent Application No. 2012/0221140, now U.S. Pat. No. 8,489,224, the disclosure of which is hereby incorporated by reference.

FIELD OF THE INVENTION
[0002] The present invention relates to systems and methodologies for automated tool path design and computer controlled machining and products produced thereby.
BACKGROUND OF THE INVENTION
[0003] Various systems and methodologies are known for automated tool path design and computer controlled machining.
SUMMARY OF THE INVENTION
[0004] The present invention seeks to provide improved systems and methodologies for automated tool path design and computer controlled machining and products produced thereby.
[0005] There is thus provided in accordance with a preferred embodiment of the present invention an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate a machined object from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, the method including defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object and calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and the number of multiple step-up cuts in the workpiece at multiple heights along the Z-axis and the areas cut in each of the multiple step-up cuts are selected so as to generally minimize the amount of workpiece material that is removed from the workpiece during the cuts while ensuring that the surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface.
[0006] There is also provided in accordance with another preferred embodiment of the present invention an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate a machined object from a workpiece

having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, the method including defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object and calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and a decision of whether or not to cut the workpiece at a given location at each height of each of the multiple step-up cuts is a function of the required non-vertical slope of the finished object at the given location.

[0007] There is further provided in accordance with yet another preferred embodiment of the present invention an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate a machined object from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, the method including defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object and calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface; and a decision as to at which height each of the multiple step-up cuts is made is a function of the required non-vertical slope of the finished object at the given height at various locations on the finished object.

[0008] Preferably, the function is a function of the smallest slope of the finished object at the given height.

[0009] There is also provided in accordance with another preferred embodiment of the present invention a method for machining a workpiece having a Z-axis, employing a computer numerically controlled milling machine, to fabricate a machined object from the workpiece, the machined object being configured to facilitate subsequent finishing into a finished object, the method including defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object, calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and the number of multiple step-up cuts in the workpiece at multiple heights along the Z-axis and the areas cut in each of the multiple step-up cuts are selected so as to generally minimize the amount of workpiece material that is removed from the workpiece during the cuts while ensuring that the surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and directing a computer controlled machine tool along the tool path.

[0010] There is further provided in accordance with yet another preferred embodiment of the present invention a method for machining a workpiece having a Z-axis, employing a computer numerically controlled milling machine, to fabricate a machined object from the workpiece, the machined object being configured to facilitate subsequent finishing into a finished object, the method including defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object, calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and a decision of whether or not to cut the workpiece at a given location at each height of each of the multiple step-up cuts is a function of the required non-vertical slope of the finished object at the given location and directing a computer controlled machine tool along the tool path.

[0011] There is even further provided in accordance with still another preferred embodiment of the present invention a method for machining a workpiece having a Z-axis, employing a computer numerically controlled milling machine, to fabricate a machined object from the workpiece, the machined object being configured to facilitate subsequent finishing into a finished object, the method including defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object, calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and a decision as to at which height each of the multiple step-up cuts is made is a function of the required non-vertical slope of the finished object at the given height at various locations on the finished object and directing a computer controlled machine tool along the tool path.

[0012] Preferably, the function is a function of the smallest slope of the finished object at the given height.

[0013] In accordance with a preferred embodiment of the present invention the calculating the tool path includes selecting the height of each of the multiple step-up cuts to be the maximum height which ensures that each of the surfaces that are cut at that height lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface.

[0014] Preferably, the calculating the tool path includes selecting whether or not to cut the workpiece at a given location at each height of each of the multiple step-up cuts. Additionally or alternatively, the calculating the tool path includes selecting the width of the cut at a given location at each height of each of the multiple step-up cuts.

[0015] In accordance with a preferred embodiment of the present invention the tool path includes at least an initial tool path portion which defines an initial cut having vertical walls followed by at least one tool path portion which further machines the vertical walls of the initial cut into a plurality of stepwise vertical walls which together define the vertical slopes at each of the plurality of surface portions which lie adjacent the initial cut and correspond to the multiple step-up cuts.

[0016] Preferably, the calculating the tool path for the computer numerically controlled milling machine includes calculating the height of a step for a collection of mutually azimuthally separated points densely distributed all along a curve representing the intersection of a step forward edge wall with a lower step floor surface. Additionally, the calculating the height of a step for a collection of mutually azimuthally separated points includes for each one of the collection of points, drawing an imaginary vertical line, parallel to the Z-axis to extend through the point and intersect at a scallop curve intersection point with the scallop surface, ascertaining the lowest height of a scallop curve intersection point corresponding to any of the collection of mutually azimuthally separated points and selecting the height for the step as being the lowest height of a scallop curve intersection point corresponding to any of the collection of mutually azimuthally separated points.

[0017] In accordance with a preferred embodiment of the present invention the automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine also includes taking an imaginary slice through the workpiece perpendicular to the Z-axis at the height for the step, ascertaining a normal distance between the a point on the imaginary vertical line at the height and the scallop surface and if the normal distance for the one of the collection of points is less than a predetermined scallop tolerance, designating the one of the collection of points as a “good to cut” point.

[0018] There is even further provided in accordance with still another preferred embodiment of the present invention an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object from a workpiece, the method including ascertaining the available spindle power of the computer numerically controlled milling machine, automatically selecting, using a computer, a maximum depth and width of cut, which are a function at least of the available spindle power of the computer numerically controlled milling machine and configuring a tool path for a tool relative to the workpiece in which the tool path includes a plurality of tool path layers whose maximum thickness and width of cut correspond to the maximum depth and width of cut.

[0019] There is still further provided in accordance with yet another preferred embodiment of the present invention a method for machining a workpiece employing a computer numerically controlled milling machine, the method including ascertaining the available spindle power of the computer numerically controlled milling machine, automatically selecting, using a computer, a maximum depth and width of cut, which are a function at least of the available spindle power of the computer numerically controlled milling machine, configuring a tool path for a tool relative to the workpiece in which the tool path includes a plurality of tool path layers whose maximum thickness and width of cut correspond to the maximum depth and width of cut and directing a computer controlled machine tool along the tool path.

[0020] Preferably, the method also includes varying at least one additional parameter of the milling machine as a function of the available spindle power. Additionally, the at least one additional parameter of the milling machine is at least one of feed speed and rpm.

[0021] There is still further provided in accordance with yet a further preferred embodiment of the present invention an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object having a relatively thin wall from a workpiece, the method including automatically selecting, using a computer, a tool path having the following characteristics: initially machining the workpiece at first maximum values of cutting depth, cutting width, cutting speed and cutting feed to have a relatively thick wall at the location of an intended relatively thin wall, reducing the height of the relatively thick wall to the intended height of the intended relatively thin wall and thereafter reducing the thickness of the thick wall by machining the workpiece at second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values.

[0022] There is also provided in accordance with another preferred embodiment of the present invention a method for machining a workpiece employing a computer numerically controlled milling machine to fabricate an object having a relatively thin wall from the workpiece, the method including automatically selecting, using a computer, a tool path having the following characteristics: initially machining the workpiece at first maximum values of cutting depth, cutting width, cutting speed and cutting feed to have a relatively thick wall at the location of an intended relatively thin wall, reducing the height of the relatively thick wall to the intended height of the intended relatively thin wall and thereafter reducing the thickness of the thick wall by machining the workpiece at second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values and directing a computer controlled machine tool along the tool path.

[0023] There is also provided in accordance with another preferred embodiment of the present invention an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object, the method including ascertaining the extent of tool overhang of a tool being used in the computer numerically controlled milling machine, automatically selecting, using a computer, a tool path which is a function of the tool overhang, the tool path having the following characteristics: for a first tool overhang selecting a tool path having first maximum values of cutting depth, cutting width, cutting speed and cutting feed and for a second tool overhang which is greater than the first tool overhang, selecting a tool path having second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values.

[0024] There is even further provided in accordance with still another preferred embodiment of the present invention a method for machining a workpiece employing a computer numerically controlled milling machine, the method including ascertaining the extent of tool overhang of a tool being used in the computer numerically controlled milling machine, automatically selecting, using a computer, a tool path which is a function of the tool overhang, the tool path having the following characteristics: for a first tool overhang selecting a tool path having first maximum values of cutting depth, cutting width, cutting speed and cutting feed and for a second tool overhang which is greater than the first tool overhang, selecting a tool path having second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values and directing the tool along the tool path.

[0025] There is further provided in accordance with yet another preferred embodiment of the present invention an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object having a semi-open region, the method including estimating, using a computer, a first machining time for machining the semi-open region using a generally trichoidal type tool path, estimating, using a computer, a second machining time for machining the semi-open region using a generally spiral type tool path and automatically selecting, using a computer, a tool path type having a shorter machining time.

[0026] There is further provided in accordance with yet another preferred embodiment of the present invention a method for machining a workpiece using a computer numerically controlled milling machine to fabricate an object having a semi-open region, the method including estimating, using a computer, a first machining time for machining the semi-open region using a generally trichoidal type tool path, estimating, using a computer, a second machining time for machining the semi-open region using a generally spiral type tool path, automatically selecting, using a computer, a tool path type having a shorter machining time and directing a computer controlled machine tool along the tool path type having a shorter machining time.

[0027] Preferably, the generally spiral type tool path is characterized in that it includes: an initial spiral type tool path portion characteristic of machining a closed region, included within the semi-open region and having a relatively thick wall separating at least one side thereof from an open edge of the semi-open region and a plurality of tool paths suitable for removal of the relatively thick wall.

[0028] In accordance with a preferred embodiment of the present invention the plurality of tool paths are suitable for cutting mutually spaced relatively narrow channels in the thick wall, thereby defining a plurality of thick wall segments and thereafter removing the plurality of thick wall segments. Additionally, the plurality of tool paths include spiral tool paths suitable for removing the plurality of thick wall segments.

[0029] There is even further provided in accordance with still another preferred embodiment of the present invention an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object having a channel open at both its ends and including an intermediate narrowest portion, the method including automatically selecting, using a computer, a tool path type having first and second tool path portions, each starting at a different open end of the channel, the first and second tool path portions meeting at the intermediate narrowest portion.

[0030] There is also provided in accordance with another preferred embodiment of the present invention a method for machining a workpiece using a computer numerically controlled milling machine to fabricate an object having a channel open at both its ends and including an intermediate narrowest portion, the method including automatically selecting, using a computer, a tool path type having first and second tool path portions, each starting at a different open end of the channel, the first and second tool path portions meeting at the intermediate narrowest portion and directing a computer controlled machine tool along the first and second tool path portions.

[0031] There is yet further provided in accordance with still another preferred embodiment of the present invention an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object which fabrication involves cutting a workpiece at least first and second different maximum depths of cut, wherein the first maximum depth of cut is greater than the second maximum depth of cut the method including automatically selecting, using a computer, at least first and second tool paths having corresponding first and second maximum values of cutting width, cutting speed and cutting feed, at least one of the second maximum values being greater than a corresponding one of the first maximum values.

[0032] There is even further provided in accordance with yet another preferred embodiment of the present invention a method for machining a workpiece using a computer numerically controlled milling machine to fabricate an object which fabrication involves cutting a workpiece at least first and second different maximum depths of cut, wherein the first maximum depth of cut is greater than the second maximum depth of cut, the method including automatically selecting, using a computer, at least first and second tool paths having corresponding first and second maximum values of cutting width, cutting speed and cutting feed, at least one of the second maximum values being greater than a corresponding one of the first maximum values and directing a computer controlled machine tool along the at least first and second tool paths.

[0033] Preferably, the automatically selecting includes adjusting the first and second maximum values of cutting width, cutting speed and cutting feed to ensure that the mechanical load experienced by a milling tool is at a generally constant optimized value.

[0034] There is still further provided in accordance with yet a further preferred embodiment of the present invention an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object, wherein fabrication of the object involves calculating multiple tool paths requiring tool repositioning therebetween along a selectable repositioning path, the method including estimating, using a computer, a first repositioning time for a first repositioning path which includes travel in a clearance plane above a workpiece, estimating, using a computer, a second repositioning time for a second repositioning path which does not include tool travel in the clearance plane and automatically selecting, using a computer, a repositioning path having a shortest repositioning time.

[0035] There is further provided in accordance with another preferred embodiment of the present invention a method for machining a workpiece using a computer numerically controlled milling machine to fabricate an object, wherein fabrication of the object involves calculating multiple tool paths requiring tool repositioning therebetween along a selectable repositioning path, the method including estimating, using a computer, a first repositioning time for a first repositioning path which includes travel in a clearance plane above a workpiece, estimating, using a computer, a second repositioning time for a second repositioning path which does not include tool travel in the clearance plane, automatically selecting, using a computer, a repositioning path having a shortest repositioning time and directing a computer controlled machine tool along the repositioning path having the shortest repositioning time.

[0036] In accordance with yet another preferred embodiment of the present invention the second repositioning path is automatically selected by the computer from among possible multiple repositioning paths which do not include tool travel in the clearance plane on the basis of shortest repositioning time. Additionally, the multiple repositioning paths include repositioning paths which require raising of the tool and repositioning paths which do not require raising of the tool.

[0037] There is also provided in accordance with another preferred embodiment of the present invention an automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate a machined object from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, the apparatus including a tool path configuration engine operative for defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object and calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and the number of multiple step-up cuts in the workpiece at multiple heights along the Z-axis and the areas cut in each of the multiple step-up cuts are selected so as to generally minimize the amount of workpiece material that is removed from the workpiece during the cuts while ensuring that the surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface.

[0038] There is further provided in accordance with yet another preferred embodiment of the present invention an automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate a machined object from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, the apparatus including a tool path configuration engine operative for defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object and calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and a decision of whether or not to cut the workpiece at a given location at each height of each of the multiple step-up cuts is a function of the required non-vertical slope of the finished object at the given location.

[0039] There is even further provided in accordance with still another preferred embodiment of the present invention an automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate a machined object from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, the apparatus including a tool path configuration engine operative for defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object and calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and a decision as to at which height each of the multiple step-up cuts is made is a function of the required non-vertical slope of the finished object at the given height at various locations on the finished object.

[0040] Preferably, the function is a function of the smallest slope of the finished object at the given height.

[0041] There is further provided in accordance with another preferred embodiment of the present invention a computer numerically controlled milling machine for fabricating a machined object from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, the computer numerically controlled milling machine including a controller operative for defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object, calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and the number of multiple step-up cuts in the workpiece at multiple heights along the Z-axis and the areas cut in each of the multiple step-up cuts are selected so as to generally minimize the amount of workpiece material that is removed from the workpiece during the cuts while ensuring that the surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and directing a computer controlled machine tool along the tool path.

[0042] There is still further provided in accordance with yet another preferred embodiment of the present invention a computer numerically controlled milling machine for fabricating a machined object from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, the computer numerically controlled milling machine including a controller operative for defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object, calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and a decision of whether or not to cut the workpiece at a given location at each height of each of the multiple step-up cuts is a function of the required non-vertical slope of the finished object at the given location and directing a computer controlled machine tool along the tool path.

[0043] There is even further provided in accordance with still another preferred embodiment of the present invention a computer numerically controlled milling machine for fabricating a machined object from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, the computer numerically controlled milling machine including a controller operative for defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object, calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and a decision as to at which height each of the multiple step-up cuts is made is a function of the required non-vertical slope of the finished object at the given height at various locations on the finished object and directing a computer controlled machine tool along the tool path.

[0044] Preferably, the function is a function of the smallest slope of the finished object at the given height.

[0045] In accordance with a preferred embodiment of the present invention the calculating the tool path includes selecting the height of each of the multiple step-up cuts to be the maximum height which ensures that each of the surfaces that are cut at that height lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface.

[0046] Preferably, the calculating the tool path includes selecting whether or not to cut the workpiece at a given location at each height of each of the multiple step-up cuts. Additionally or alternatively, the calculating the tool path includes selecting the width of the cut at a given location at each height of each of the multiple step-up cuts.

[0047] In accordance with a preferred embodiment of the present invention the tool path includes at least an initial tool path portion which defines an initial cut having vertical walls followed by at least one tool path portion which further machines the vertical walls of the initial cut into a plurality of stepwise vertical walls which together define the vertical slopes at each of the plurality of surface portions which lie adjacent the initial cut and correspond to the multiple step-up cuts.

[0048] Preferably, the calculating the tool path for the computer numerically controlled milling machine includes calculating the height of a step for a collection of mutually azimuthally separated points densely distributed all along a curve representing the intersection of a step forward edge wall with a lower step floor surface. Additionally, the calculating the height of a step for a collection of mutually azimuthally separated points includes for each one of the collection of points, drawing an imaginary vertical line, parallel to the Z-axis to extend through the point and intersect at a scallop curve intersection point with the scallop surface, ascertaining the lowest height of a scallop curve intersection point corresponding to any of the collection of mutually azimuthally separated points and selecting the height for the step as being the lowest height of a scallop curve intersection point corresponding to any of the collection of mutually azimuthally separated points.

[0049] In accordance with a preferred embodiment of the present invention the calculating the tool path for the computer numerically controlled milling machine also includes taking an imaginary slice through the workpiece perpendicular to the Z-axis at the height for the step, ascertaining a normal distance between the a point on the imaginary vertical line at the height and the scallop surface and if the normal distance for the one of the collection of points is less than a predetermined scallop tolerance, designating the one of the collection of points as a “good to cut” point.

[0050] There is even further provided in accordance with still another preferred embodiment of the present invention an automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate an object from a workpiece, the apparatus including a tool path configuration engine operative for ascertaining the available spindle power of the computer numerically controlled milling machine, automatically selecting, using a computer, a maximum depth and width of cut, which are a function at least of the available spindle power of the computer numerically controlled milling machine and configuring a tool path for a tool relative to the workpiece in which the tool path includes a plurality of tool path layers whose maximum thickness and width of cut correspond to the maximum depth and width of cut.

[0051] There is also provided in accordance with another preferred embodiment of the present invention a computer numerically controlled milling machine including a controller operative for ascertaining the available spindle power of the computer numerically controlled milling machine, automatically selecting a maximum depth and width of cut, which are a function at least of the available spindle power of the computer numerically controlled milling machine, configuring a tool path for a tool relative to a workpiece in which the tool path includes a plurality of tool path layers whose maximum thickness and width of cut correspond to the maximum depth and width of cut and directing a computer controlled machine tool along the tool path.

[0052] Preferably, the automatically selecting also includes varying at least one additional parameter of the milling machine as a function of the available spindle power. Additionally, the at least one additional parameter of the milling machine is at least one of feed speed and rpm.

[0053] There is still further provided in accordance with yet a further preferred embodiment of the present invention an automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate an object having a relatively thin wall from a workpiece, the apparatus including a tool path configuration engine operative for automatically selecting, using a computer, a tool path having the following characteristics: initially machining the workpiece at first maximum values of cutting depth, cutting width, cutting speed and cutting feed to have a relatively thick wall at the location of an intended relatively thin wall, reducing the height of the relatively thick wall to the intended height of the intended relatively thin wall; and thereafter reducing the thickness of the thick wall by machining the workpiece at second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values.

[0054] There is further provided in accordance with yet another preferred embodiment of the present invention a computer numerically controlled milling machine for fabricating an object having a relatively thin wall from a workpiece, the computer numerically controlled milling machine including a controller operative for automatically selecting a tool path having the following characteristics: initially machining the workpiece at first maximum values of cutting depth, cutting width, cutting speed and cutting feed to have a relatively thick wall at the location of an intended relatively thin wall, reducing the height of the relatively thick wall to the intended height of the intended relatively thin wall and thereafter reducing the thickness of the thick wall by machining the workpiece at second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values and directing a computer controlled machine tool along the tool path.

[0055] There is also provided in accordance with another preferred embodiment of the present invention an automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate an object, the apparatus including a tool path configuration engine operative for ascertaining the extent of tool overhang of a tool being used in the computer numerically controlled milling machine, automatically selecting, using a computer, a tool path which is a function of the tool overhang, the tool path having the following characteristics: for a first tool overhang selecting a tool path having first maximum values of cutting depth, cutting width, cutting speed and cutting feed and for a second tool overhang which is greater than the first tool overhang, selecting a tool path having second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values.

[0056] There is further provided in accordance with yet another preferred embodiment of the present invention a computer numerically controlled milling machine for machining a workpiece, the computer numerically controlled milling machine including a controller operative for ascertaining the extent of tool overhang of a tool being used in the computer numerically controlled milling machine, automatically selecting a tool path which is a function of the tool overhang, the tool path having the following characteristics: for a first tool overhang selecting a tool path having first maximum values of cutting depth, cutting width, cutting speed and cutting feed, for a second tool overhang which is greater than the first tool overhang, selecting a tool path having second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values and directing the tool along the tool path.

[0057] There is further provided in accordance with yet another preferred embodiment of the present invention an automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate an object having a semi-open region, the apparatus including a tool path configuration engine operative for estimating, using a computer, a first machining time for machining the semi-open region using a generally trichoidal type tool path, estimating, using a computer, a second machining time for machining the semi-open region using a generally spiral type tool path and automatically selecting, using a computer, a tool path type having a shorter machining time.

[0058] There is still further provided in accordance with yet a further preferred embodiment of the present invention a computer numerically controlled milling machine for fabricating an object having a semi-open region from a workpiece, the computer numerically controlled milling machine including a controller operative for estimating a first machining time for machining the semi-open region using a generally trichoidal type tool path, estimating a second machining time for machining the semi-open region using a generally spiral type tool path, automatically selecting a tool path type having a shorter machining time and directing a computer controlled machine tool along the tool path type having a shorter machining time.

[0059] Preferably, the generally spiral type tool path is characterized in that it includes: an initial spiral type tool path portion characteristic of machining a closed region, included within the semi-open region and having a relatively thick wall separating at least one side thereof from an open edge of the semi-open region and a plurality of tool paths suitable for removal of the relatively thick wall.

[0060] In accordance with a preferred embodiment of the present invention the plurality of tool paths are suitable for cutting mutually spaced relatively narrow channels in the thick wall, thereby

defining a plurality of thick wall segments and thereafter removing the plurality of thick wall segments. Additionally, the plurality of tool paths include spiral tool paths suitable for removing the plurality of thick wall segments.
[0061] There is even further provided in accordance with still another preferred embodiment of the present invention an automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate an object having a channel open at both its ends and including an intermediate narrowest portion, the apparatus including a tool path configuration engine operative for automatically selecting, using a computer, a tool path type having first and second tool path portions, each starting at a different open end of the channel, the first and second tool path portions meeting at the intermediate narrowest portion.
[0062] There is also provided in accordance with still another preferred embodiment of the present invention a computer numerically controlled milling machine for machining a workpiece to fabricate an object having a channel open at both its ends and including an intermediate narrowest portion, the computer numerically controlled milling machine including a controller operative for automatically selecting a tool path type having first and second tool path portions, each starting at a different open end of the channel, the first and second tool path portions meeting at the intermediate narrowest portion and directing a computer controlled machine tool along the first and second tool path portions.
[0063] There is yet further provided in accordance with still another preferred embodiment of the present invention an automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate an object which fabrication involves cutting a workpiece at least first and second different maximum depths of cut, wherein the first maximum depth of cut is greater than the second maximum depth of cut the apparatus including a tool path configuration engine operative for automatically selecting, using a computer, at least first and second tool paths having corresponding first and second maximum values of cutting width, cutting speed and cutting feed, at least one of the second maximum values being greater than a corresponding one of the first maximum values.
[0064] There is even further provided in accordance with yet another preferred embodiment of the present invention a computer numerically controlled milling machine for machining a workpiece to fabricate an object which fabrication involves cutting a workpiece at least first and second different maximum depths of cut, wherein the first maximum depth of cut is greater than the second maximum depth of cut, the computer numerically controlled milling machine including a controller operative for automatically selecting at least first and second tool paths having corresponding first and second maximum values of cutting width, cutting speed and cutting feed, at least one of the second maximum values being greater than a corresponding one of the first maximum values and directing a computer controlled machine tool along the at least first and second tool paths.
[0065] Preferably, the automatically selecting includes adjusting the first and second maximum values of cutting width, cutting speed and cutting feed to ensure that the mechanical load experienced by a milling tool is at a generally constant optimized value.
[0066] There is still further provided in accordance with yet a further preferred embodiment of the present invention an automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate an object, wherein fabrication of the object involves calculating multiple tool paths requiring tool repositioning therebetween along a selectable repositioning path, the apparatus including a tool path configuration engine operative for estimating, using a computer, a first repositioning time for a first repositioning path which includes travel in a clearance plane above a workpiece, estimating, using a computer, a second repositioning time for a second repositioning path which does not include tool travel in the clearance plane and automatically selecting, using a computer, a repositioning path having a shortest repositioning time.
[0067] There is further provided in accordance with another preferred embodiment of the present

invention a computer numerically controlled milling machine for machining a workpiece to fabricate an object, wherein fabrication of the object involves calculating multiple tool paths requiring tool repositioning therebetween along a selectable repositioning path, the computer numerically controlled milling machine including a controller operative for estimating a first repositioning time for a first repositioning path which includes travel in a clearance plane above a workpiece, estimating a second repositioning time for a second repositioning path which does not include tool travel in the clearance plane, automatically selecting a repositioning path having a shortest repositioning time and directing a computer controlled machine tool along the repositioning path having the shortest repositioning time.

[0068] In accordance with a preferred embodiment of the present invention the second repositioning path is automatically selected by the computer from among possible multiple repositioning paths which do not include tool travel in the clearance plane on the basis of shortest repositioning time. Additionally, the multiple repositioning paths include repositioning paths which require raising of the tool and repositioning paths which do not require raising of the tool.

[0069] There is also provided in accordance with another preferred embodiment of the present invention a machined object fabricated from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, using a computer numerically controlled milling machine by defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object, calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and the number of multiple step-up cuts in the workpiece at multiple heights along the Z-axis and the areas cut in each of the multiple step-up cuts are selected so as to generally minimize the amount of workpiece material that is removed from the workpiece during the cuts while ensuring that the surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and directing a computer controlled machine tool along the tool path.

[0070] There is further provided in accordance with still another preferred embodiment of the present invention a machined object fabricated from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, using a computer numerically controlled milling machine by defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object, calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and a decision of whether or not to cut the workpiece at a given location at each height of each of the multiple step-up cuts is a function of the required non-vertical slope of the finished object at the given location and directing a computer controlled machine tool along the tool path.

[0071] There is yet further provided in accordance with another preferred embodiment of the present invention a machined object fabricated from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object, using a computer numerically controlled milling machine by defining a surface of the finished object, defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object, defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object, calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface and a decision as to at which height each of the multiple step-up cuts is made is a function of the required non-vertical slope of the finished object at the given height at various locations on the finished object and directing a computer controlled machine tool along the tool path.

[0072] Preferably, the function is a function of the smallest slope of the finished object at the given height.

[0073] In accordance with a preferred embodiment of the present invention the calculating the tool path includes selecting the height of each of the multiple step-up cuts to be the maximum height which ensures that each of the surfaces that are cut at that height lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface.

[0074] Preferably, the calculating the tool path includes selecting whether or not to cut the workpiece at a given location at each height of each of the multiple step-up cuts. In accordance with a preferred embodiment of the present invention the calculating the tool path includes selecting the width of the cut at a given location at each height of each of the multiple step-up cuts.

[0075] Preferably, the tool path includes at least an initial tool path portion which defines an initial cut having vertical walls followed by at least one tool path portion which further machines the vertical walls of the initial cut into a plurality of stepwise vertical walls which together define the vertical slopes at each of the plurality of surface portions which lie adjacent the initial cut and correspond to the multiple step-up cuts.

[0076] Preferably, the calculating the tool path for the computer numerically controlled milling machine includes calculating the height of a step for a collection of mutually azimuthally separated points densely distributed all along a curve representing the intersection of a step forward edge wall with a lower step floor surface. In accordance with a preferred embodiment of the present invention the calculating the height of a step for a collection of mutually azimuthally separated points includes for each one of the collection of points, drawing an imaginary vertical line, parallel to the Z-axis to extend through the point and intersect at a scallop curve intersection point with the scallop surface, ascertaining the lowest height of a scallop curve intersection point corresponding to any of the collection of mutually azimuthally separated points and selecting the height for the step as being the lowest height of a scallop curve intersection point corresponding to any of the collection of mutually azimuthally separated points.

[0077] Preferably, the calculating the tool path also includes taking an imaginary slice through the workpiece perpendicular to the Z-axis at the height for the step, ascertaining a normal distance between the a point on the imaginary vertical line at the height and the scallop surface and if the normal distance for the one of the collection of points is less than a predetermined scallop tolerance, designating the one of the collection of points as a “good to cut” point.

[0078] There is even further provided in accordance with yet another preferred embodiment of the present invention a machined object fabricated from a workpiece using a computer numerically controlled milling machine by ascertaining the available spindle power of the computer numerically controlled milling machine, automatically selecting a maximum depth and width of cut, which are a function at least of the available spindle power of the computer numerically controlled milling machine, configuring a tool path for a tool relative to the workpiece in which the tool path includes a plurality of tool path layers whose maximum thickness and width of cut correspond to the maximum depth and width of cut and directing a computer controlled machine tool along the tool path.

[0079] Preferably, the automatically selecting also includes varying at least one additional parameter of the milling machine as a function of the available spindle power. In accordance with a preferred embodiment of the present invention the at least one additional parameter of the milling machine is at least one of feed speed and rpm.

[0080] [**00**.] There is still further provided in accordance with still another preferred embodiment of the present invention a machined object having a relatively thin wall fabricated from a workpiece using a computer numerically controlled milling machine by automatically selecting a tool path having the following characteristics: initially machining the workpiece at first maximum values of cutting depth, cutting width, cutting speed and cutting feed to have a relatively thick wall at the location of an intended relatively thin wall, reducing the height of the relatively thick wall to the intended height of the intended relatively thin wall and thereafter reducing the thickness of the thick wall by machining the workpiece at second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values and directing a computer controlled machine tool along the tool path.

[0081] There is also provided in accordance with another preferred embodiment of the present invention a machined object machined from a workpiece using a computer numerically controlled milling machine by ascertaining the extent of tool overhang of a tool being used in the computer numerically controlled milling machine, automatically selecting a tool path which is a function of the tool overhang, the tool path having the following characteristics: for a first tool overhang selecting a tool path having first maximum values of cutting depth, cutting width, cutting speed and cutting feed; for a second tool overhang which is greater than the first tool overhang, selecting a tool path having second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values and directing the tool along the tool path.

[0082] There is further provided in accordance with yet another preferred embodiment of the present invention a machined object having a semi-open region fabricated from a workpiece using a computer numerically controlled milling machine by estimating a first machining time for machining the semi-open region using a generally trichoidal type tool path, estimating a second machining time for machining the semi-open region using a generally spiral type tool path, automatically selecting a tool path type having a shorter machining time and directing a computer controlled machine tool along the tool path type having a shorter machining time.

[0083] Preferably, the generally spiral type tool path is characterized in that it includes an initial spiral type tool path portion characteristic of machining a closed region, included within the semi-open region and having a relatively thick wall separating at least one side thereof from an open edge of the semi-open region and a plurality of tool paths suitable for removal of the relatively thick wall.

[0084] In accordance with a preferred embodiment of the present invention the plurality of tool paths are suitable for cutting mutually spaced relatively narrow channels in the thick wall, thereby defining a plurality of thick wall segments and thereafter removing the plurality of thick wall segments. Additionally, the plurality of tool paths include spiral tool paths suitable for removing the plurality of thick wall segments.

[0085] There is still further provided in accordance with still another preferred embodiment of the present invention a machined object having a channel open at both its ends and including an intermediate narrowest portion fabricated using a computer numerically controlled milling machine

by automatically selecting a tool path type having first and second tool path portions, each starting at a different open end of the channel, the first and second tool path portions meeting at the intermediate narrowest portion and directing a computer controlled machine tool along the first and second tool path portions.
[0086] There is even further provided in accordance with yet a further preferred embodiment of the present invention a machined object, the fabrication of which involves cutting a workpiece at least first and second different maximum depths of cut, wherein the first maximum depth of cut is greater than the second maximum depth of cut, fabricated using a computer numerically controlled milling machine by automatically selecting at least first and second tool paths having corresponding first and second maximum values of cutting width, cutting speed and cutting feed, at least one of the second maximum values being greater than a corresponding one of the first maximum values and directing a computer controlled machine tool along the at least first and second tool paths.
[0087] In accordance with a preferred embodiment of the present invention the automatically selecting includes adjusting the first and second maximum values of cutting width, cutting speed and cutting feed to ensure that the mechanical load experienced by a milling tool is at a generally constant optimized value.
[0088] There is also provided in accordance with yet another preferred embodiment of the present invention a machined object, the fabrication of which involves calculating multiple tool paths requiring tool repositioning therebetween along a selectable repositioning path, fabricated using a computer numerically controlled milling machine by estimating a first repositioning time for a first repositioning path which includes travel in a clearance plane above a workpiece, estimating a second repositioning time for a second repositioning path which does not include tool travel in the clearance plane, automatically selecting a repositioning path having a shortest repositioning time and directing a computer controlled machine tool along the repositioning path having the shortest repositioning time.
[0089] Preferably, the second repositioning path is automatically selected by the computer from among possible multiple repositioning paths which do not include tool travel in the clearance plane on the basis of shortest repositioning time. Additionally, the multiple repositioning paths include repositioning paths which require raising of the tool and repositioning paths which do not require raising of the tool.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0090] The present invention will be understood and appreciated from the following detailed description, taken in conjunction with the drawings in which:
[0091] FIGS. **1**A-**1**S-**2** are together a series of simplified illustrations which are helpful in understanding the invention;
[0092] FIGS. **2**A-**2**L-**2** are together another series of simplified illustrations which are helpful in understanding the invention;
[0093] FIGS. **3**A-**3**D are simplified screen shots illustrating some aspects of the present invention;
[0094] FIGS. **4**A and **4**B are simplified illustrations of details of functionality illustrated more generally in certain ones of FIGS. **1**A-**1**S-**2** and FIGS. **2**A-**2**L-**2**;
[0095] FIG. **5** is a simplified illustration of a workpiece, a machined object formed from the workpiece in accordance with a preferred embodiment of the present invention, and a finished object to be produced from the machined object;
[0096] FIG. **6** is a simplified annotated pictorial illustration of the machined object of FIG. **5** showing, in sectional enlargements, finished object surface, offset surface and scallop surface

profiles at two mutually azimuthally separated locations in accordance with a preferred embodiment of the present invention;

[0097] FIG. **7** is a simplified pictorial illustration of the workpiece of FIG. **5** showing an initial two-dimensional deep cut therein and also showing, in enlargements the cut superimposed over the corresponding enlargements of FIG. **6**;

[0098] FIG. **8** is a simplified top view illustration of the workpiece of FIG. **5** following the initial cut illustrated in FIG. **7** and showing a tool path preferably employed to achieve this cut in accordance with a preferred embodiment of the present invention;

[0099] FIGS. **9**A, **9**B and **9**C are simplified sectional illustrations of the workpiece of FIGS. **7** and **8** superimposed over the corresponding annotated sectional illustrations in the enlargements of FIG. **6** and respectively showing elements of the calculation of first, second and third step-up cuts in accordance with a preferred embodiment of the present invention;

[0100] FIGS. **10**A and **10**B are simplified sectional illustrations of the workpiece of FIGS. **7-9**A showing a first step-up cut in accordance with a preferred embodiment of the present invention;

[0101] FIG. **10**C is a simplified top view illustration of a portion of the workpiece of FIGS. **7-9**A following the step-up cut illustrated in FIGS. **10**A & **10**B in accordance with a preferred embodiment of the present invention;

[0102] FIGS. **11**A and **11**B are simplified sectional illustrations of the workpiece of FIGS. **7-9**B showing a second step-up cut in accordance with a preferred embodiment of the present invention;

[0103] FIG. **11**C is a simplified top view illustration of a portion of the workpiece of FIGS. **7-9**B following the step-up cut illustrated in FIGS. **11**A & **11**B in accordance with a preferred embodiment of the present invention;

[0104] FIGS. **12**A and **12**B are simplified sectional illustrations of the workpiece of FIGS. **7-9**C showing a first step-up cut in accordance with a preferred embodiment of the present invention;

[0105] FIG. **12**C is a simplified top view illustration of a portion of the workpiece of FIGS. **7-9**C following the step-up cut illustrated in FIGS. **12**A & **12**B in accordance with a preferred embodiment of the present invention;

[0106] FIG. **13** is a simplified partially symbolic, partially pictorial illustration of exemplary selection of axial depth of cut and stepover as a function of available spindle power in accordance with a preferred embodiment of the present invention;

[0107] FIG. **14** is a simplified pictorial illustration of tool paths for machining objects having thin walls in accordance with a preferred embodiment of the present invention;

[0108] FIG. **15** is a simplified partially symbolic, partially pictorial illustration of varying stepover as a function of tool overhang in accordance with a preferred embodiment of the present invention;

[0109] FIG. **16** is a simplified pictorial illustration of machining functionality in accordance with another preferred embodiment of the present invention;

[0110] FIG. **17** is a simplified pictorial illustration of tool paths for machining objects having an hourglass-shaped channel in accordance with a preferred embodiment of the present invention;

[0111] FIG. **18** is a simplified pictorial illustration of tool paths for machining objects which are calculated based on optimal cutting conditions for a maximum cutting depth and tool paths for cutting portions which involve cutting at less than the maximum cutting depth are calculated based on modified cutting conditions optimized for cutting at a depth less than the maximum cutting depth in accordance with a preferred embodiment of the present invention; and

[0112] FIGS. **19**A, **19**B and **19**C are simplified illustrations of three alternative tool repositioning moves from the end of one tool path to the beginning of a subsequent tool path in the same pocket in accordance with a preferred embodiment of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0113] The present invention relates to various aspects of an automated computer-implemented method for generating commands for controlling a computer numerical controlled (CNC) machine to fabricate an object from a stock material, various aspects of a method for machining the stock material which employs the above commands, automated computer-implemented apparatus for generating the above commands, a numerically-controlled machine operative to fabricate an object from a stock material by using the above commands, and an object fabricated by using the above commands.

[0114] The invention, in its various aspects, is described hereinbelow with respect to a series of drawings, which initially illustrate an example of an object to be fabricated, a simulated overlay of the object on a stock material to be machined and sequences of machining steps that are produced by commands generated in accordance with the present invention. It is appreciated that although sequential machining steps are illustrated, the invention is not limited to a machining method but extends as noted above to the generation of the commands, the apparatus, which generates them, the apparatus which carries them out and to the result produced thereby.

[0115] The term “calculation” is used throughout to refer to the generation of commands which produce sequences of machining steps to be employed in the machining of a particular region of the stock material. The definitions “calculate”, “calculation” and calculation are of corresponding meaning.

[0116] FIGS. **1**A and **1**B are respective pictorial and top view illustrations of an object **100** which is an example of objects that can be fabricated in accordance with the present invention. The configuration of the object **100** is selected to illustrate various particular features of the present invention. It is noted that any suitable three-dimensional object that can be machined by a conventional 3-axis CNC machine tool may be fabricated in accordance with a preferred embodiment of the present invention.

[0117] As seen in FIGS. **1**A & **1**B, the object **100** is seen to have a generally planar base portion **102** from which five protrusions, here designated by reference numerals **104**, **106**, **108**, **110** and **112** extend. FIG. **1**C shows stock material **114** overlaid by an outline of object **100**.

[0118] In accordance with a preferred embodiment of the present invention, a tool path designer, using the automated computer-implemented method for generating commands for controlling a computer numerical controlled machine of the present invention, accesses a CAD drawing of the object **100** in a standard CAD format, such as SOLIDWORKS®. He selects a specific machine tool to be used in fabrication of the object **100** from a menu and selects a specific rotating cutting tool to carry out each machining function required to fabricate the object.

[0119] For the sake of simplicity, the illustrated object **100** is chosen to be an object that can be fabricated by a single machining function, it being appreciated that the applicability of the present invention is not limited to objects which can be fabricated by a single machining function.

[0120] The tool path designer then defines the geometry of the stock material to be used in fabrication of the object **100**. This may be done automatically by the automated computer-implemented apparatus of the present invention or manually by the tool path designer. The tool path designer then specifies the material which constitutes the stock material, for example, INCONEL® 718. The present invention utilizes the choice of machine tool, rotating cutting tool and the material by the tool path designer to calculate various operational parameters, based on characteristics of the machine tool, rotating cutting tool and material.

[0121] In accordance with a preferred embodiment of the present invention, a series of display screens are employed to provide a display for the tool path designer, indicating the various operational parameters, such as minimum and maximum surface cutting speed, minimum and maximum chip thickness, minimum and maximum feed speed, minimum and maximum spindle rotational speed, minimum and maximum engagement angles between the rotating cutting tool and the workpiece, axial depth of cut, machining aggressiveness level. An example of such a series of display screens appears in FIGS. **3**A-**3**D.

[0122] The tool path designer is given limited latitude in changing some of the parameters, such as particularly, the machining aggressiveness level. Preferably, the tool path designer may also instruct the system to select parameters for which, for example, optimization of machining time, wear inflicted on the cutting tool, machining cost or any combination thereof is achieved. It is appreciated that although for some of the operational parameters described hereinabove a range of values is displayed to the tool path designer, the present invention also calculates an optimal operational value for all of the operational parameters to be employed.

[0123] Once all of the parameters appearing on the screen, such as the display screens of FIGS. **3**A-**3**D, are finalized, a tool path for machining the workpiece is calculated in accordance with a preferred embodiment of the present invention. The calculation of a tool path in accordance with a preferred embodiment of the present invention is described hereinbelow with reference to FIGS. **1**A-**1**S-**2** which illustrate the actual progression of tool path in stock material **114**.

[0124] It is a particular feature of the present invention that the tool path is calculated recursively, whereby initially a first tool path segment of the tool path is calculated for an initial region of the workpiece, and thereafter a subsequent sequential tool path segment of the tool path is similarly calculated for an initial region of a remaining region of the workpiece. Additional subsequent sequential tool path segments are similarly calculated, until a tool path for machining the entire workpiece to the desired object has been calculated.

[0125] Initially, a first cross section of the stock material having the outline of the object **100** overlaid thereon and having a depth equal to the designated axial depth of cut is calculated. This cross section is illustrated schematically in FIG. **1**D and is designated by reference numeral **116**. Cross section **116** is characterized as having an external boundary **118** and a plurality of islands **105**, **107**, **109**, **111** and **113** respectively corresponding to the cross sections of protrusions **104**, **106**, **108**, **110** and **112** at the depth of cross section **116**. It is appreciated that islands **105**, **107**, **109**, **111** and **113** are offset externally to the cross sections of protrusions **104**, **106**, **108**, **110** and **112** by a distance which is generally a bit larger than the radius of the rotating cutting tool, thereby when machining a tool path which circumvents the islands, a narrow finishing width remains to be finish machined at a later stage.

[0126] It is appreciated that the axial depth of cross section **116** constitutes a first step down which is a first phase in the machining of object **100**. Throughout, the term “step down” is used to describe a single machining phase at a constant depth. As shown in FIG. **1**C, the complete machining of object **100** requires two additional step downs corresponding to cross sections **119** and **120**. Therefore, subsequent to the calculation of cross section **116**, a second step down and thereafter a third step down are calculated, corresponding to cross sections **119** and **120**. Preferably, the vertical distance between subsequent step downs is generally between 1 and 4 times the diameter of the rotating cutting tool.

[0127] In accordance with a preferred embodiment of the present invention, a machining region is initially automatically identified in cross section **116**. There are preferably three types of machining regions which are classified by the characteristics of their exterior boundaries. Throughout, a segment of the boundary of a region through which the region can be reached by a rotating cutting tool from the outside of the region by horizontal progression of the rotating cutting tool is termed an “open edge”. All other boundary segments are termed throughout as “closed edges”.

[0128] The three types of machining regions are classified as follows: [0129] Type I—an open region characterized in that the entire exterior boundary of the region consists solely of open edges; [0130] Type II—a semi-open region characterized in that the exterior boundary of the region consists of both open edges and closed edges; [0131] Type III—a closed region characterized in that the entire exterior boundary of the region consists solely of closed edges;

[0132] Preferably, a tool path to be employed in machining a region is calculated to comprise one or more tool path segments, wherein each tool path segment is one of a converging spiral tool path segment, a trochoidal-like tool path segment and a diverging spiral tool path segment. Generally, a converging spiral tool path segment is preferred when machining a Type I region, a trochoidal-like tool path segment is preferred when machining a Type II region, and a diverging spiral tool path is preferred when machining a Type III region.

[0133] The term "trochoidal-like" is used throughout to mean a trochoidal tool path or a modification thereof that retains a curved cutting path and a return path which could be either curved or generally straight.
[0134] As known to persons skilled in the art, the machining of spiral tool path segments is generally more efficient with respect to the amount of material removed per unit of time than the machining of trochoidal-like tool path segments for generally similar average stepovers. Therefore, the present invention seeks to maximize the area to be machined by spiral tool path segments.
[0135] A converging spiral tool path segment calculated to machine a Type I region preferably is a tool path segment which spirals inwardly from an external boundary of the region to an internal contour. The internal contour is preferably calculated as follows:
[0136] In a case where there are no islands within the external boundary of the Type I region, the internal contour is preferably calculated to be a small circle having a radius which is generally smaller than the radius of the cutting tool, and which is centered around the center of area of the region;
[0137] In a case where there is one island within the external boundary of the Type I region, and the shortest distance between the one island and the external boundary of the Type I region is longer than a selected fraction of the diameter of the rotating cutting tool, the internal contour is preferably calculated to be generally alongside the external boundary of the island; and
[0138] In a case where: [0139] there is one island within the external boundary of the Type I region and the shortest distance between the single island and the external boundary of the Type I region is shorter than a selected fraction of the diameter of the rotating cutting tool; or [0140] there is more than one island within the external boundary of the Type I region
the internal contour is preferably calculated to be a contour which is offset interiorly to the external boundary of the region by a distance which is generally equal to 1.5 radii of the rotating cutting tool.
[0141] Once the internal contour is calculated, it is automatically verified that the internal contour does not self intersect. In a case where the internal contour does self intersect at one or more locations, preferably a bottleneck is identified in the vicinity of each such self intersection. If the bottleneck does not overlap with an island, a separating channel is preferably calculated at each such bottleneck. A separating channel preferably divides the region into two Type I regions which can be machined independently of each other by separate converging spiral tool path segments. If the bottleneck does overlap with an island, the internal contour is preferably recalculated to be offset interiorly to the external boundary by generally half of the original offset. This process is repeated until an internal contour which does not self-intersect is calculated.
[0142] It is a particular feature of the present invention that a converging spiral tool path segment which spirals inwardly from an external boundary of a region to an internal contour is calculated to be a "morphing spiral". The term "morphing spiral" is used throughout to mean a spiral tool path segment which gradually morphs the geometrical shape of one boundary or contour to the geometrical shape of a second boundary or contour as the spiral tool path segment spirals therebetween. While various methods of morphing are known to persons skilled in the art, the present invention seeks to implement particular methods of morphing in accordance with preferred embodiments of the present invention, as described hereinbelow.
[0143] It is another particular feature of the present invention that the engagement angle of the cutting tool employed throughout the tool path segment is not fixed, but rather may vary between the predetermined minimum and maximum engagement angles over the course of the tool path segment. This varying of the engagement angle allows for varying stepovers over the course of the tool path segment, and thereby enables the tool path segment to morph between two generally dissimilar geometrical shapes. The term "stepover" is used throughout to designate the distance between sequential loops of a spiral tool path segment. It is appreciated that the cutting tool efficiency which is achieved by employing a morphing spiral tool path segment is generally

significantly greater than the cutting tool efficiency which is achieved by employing a trochoidal-like tool path segment. It is also appreciated that where appropriate, an engagement angle which is generally close to the maximum engagement angle is preferred.

[0144] While it is appreciated that employing varying engagement angles over the course of a tool path segment may have a negative impact of increasing the wear of the cutting tool due to the varying mechanical load on the cutting tool and to chip thinning, it is a particular feature of the present invention that this negative impact is generally compensated for by automatically dynamically adjusting the feed velocity to correspond to the varying engagement angle. It is another particular feature of the present invention that the engagement angle is varied gradually over the course of the tool path segment, thereby preventing sudden and sharp changes in cutting tool load, and thereby further reducing excess wear of the cutting tool.

[0145] Returning now to the calculation of a converging spiral tool path segment employed to machine a Type I region, once an internal contour has been calculated, the number of loops to be included in a converging spiral tool path segment which spirals inwardly from the external boundary of the region to the internal contour is calculated preferably as illustrated in FIG. **4**A.

[0146] As shown in FIG. **4**A, a plurality of bridges **500** of a predefined density are each stretched from the internal contour **502** to the external boundary **504**. A bridge point **506** of each of bridges **500** is initially defined as the point of intersection of bridge **500** with external boundary **504**. The length of the shortest bridge divided by the minimum stepover is generally equal to the maximum number of loops that can be included in the spiral tool path segment. The length of the longest bridge divided by the maximum stepover is generally equal to the minimum number of loops which must be included in the spiral tool path. As described hereinabove, minimum and maximum engagement angles are determined based on information provided by the tool path designer, which angles determine the minimum and maximum stepover of the spiral tool path segment.

[0147] It is appreciated that the furthest distance, in any direction, from internal contour **502** which can be machined by a converging spiral tool path segment is the number of loops included in the converging spiral tool path segment multiplied by the maximum stepover. Areas between internal contour **502** and external boundary **504** beyond this furthest distance from the internal contour cannot be machined by the converging spiral tool path segment, and are therefore preferably machined by clipping prior to the machining of the converging spiral tool path segment. Throughout, the term “clipping” is used to define the calculation of machining of areas of a region which cannot be machined by an optimal spiral tool path segment. Typically, clipped areas are machined either by a trochoidal-like tool path segment, before the machining of the spiral tool path segment, or by machining a separating channel which separates the clipped area from the remainder of the region and by subsequently machining the separated clipped area separately by a spiral tool path segment.

[0148] Throughout, a parameter ‘n’ will be used to designate a possible number of loops to be included in a spiral tool path segment, wherein n is a number between the minimum number of loops which must be included in the spiral tool path segment and the maximum number of loops that can be included in the spiral tool path segment.

[0149] For each possible value of n, a first work time for a first machining method needed to machine the area between external boundary **504** and internal contour **502** is calculated by summing the time needed to machine the spiral tool path segment and the time needed to machine all clipped areas which were identified between external boundary **504** and internal contour **502** as described hereinabove. The optimal number of loops to be included in the spiral tool path segment is chosen to be the value of n for which the first calculated work time is the shortest.

[0150] In a case where the internal contour is calculated to be a small circle which is centered around the center of area of the region, a second work time for a second machining method is calculated by summing the work time needed to machine a separating channel extending along the shortest bridge connecting the external boundary to the internal contour, further extending through the small circle and then further extending along an opposite bridge up to an opposite segment of the external boundary, thus dividing the region into two independent Type I regions, and the work time needed to machine the two independent Type I regions. In a case where the second work time is shorter than the first work time, the second machining method is preferred over the first machining method.

[0151] Once the optimal number of loops to be included in the converging spiral tool path segment is chosen, clipped areas and tool paths for their removal are calculated as described hereinabove. Subsequently, a new external boundary defined by the clipped areas is calculated and all bridge points are updated accordingly to be located on the new external boundary. Thereafter, the actual path of the spiral tool path segment is calculated, as follows:

[0152] Initially, the bridge point **510** of a first bridge **512** is preferably selected as a first spiral point of spiral tool path segment **514**. First bridge **512** is preferably selected to minimize the time required to move the cutting tool from its previous position. A possible second spiral point of spiral tool path segment **514** is calculated as a point on a second bridge **516**, immediately adjacent to first bridge **512** in a climbing direction of the cutting tool from first bridge **512**, which point is distanced from bridge point **517** of second bridge **516** along second bridge **516** by the length of second bridge **516** divided by the remaining number of loops to be included in tool path segment **514**.

[0153] For the possible second spiral point, the engagement angle at which the cutting tool will engage the material by following the spiral tool path segment **514** from first spiral point **510** to the possible second spiral point is calculated. In a case where the calculated engagement angle is between the predetermined minimum and maximum engagement angles, the possible second spiral point is chosen as the second spiral point **518**, and a new linear subsegment **520** between first spiral point **510** and second spiral point **518** is added to spiral tool path segment **514**.

[0154] In a case where the engagement angle is less than the predetermined minimum engagement angle, a binary search for a second spiral point for which the calculated engagement angle is generally equal to the predetermined minimum engagement angle is performed. The binary search is performed between the possible second spiral point and a point on second bridge **516** distanced from bridge point **517** of second bridge **516** by the maximum stepover. Once a second spiral point **518** is found, a new linear subsegment **520** between first spiral point **510** and second spiral point **518** is added to spiral tool path segment **514**.

[0155] In a case where the engagement angle is greater than the predetermined maximum engagement angle, a binary search for a second spiral point for which the calculated engagement angle is generally equal to the predetermined maximum engagement angle is performed. The binary search is performed between bridge point **517** of the second bridge **516** and the possible second spiral point. Once a second spiral point **518** is found, a new linear subsegment **520** between first spiral point **510** and second spiral point **518** is added to spiral tool path segment **514**.

[0156] In a case where new linear subsegment **520** intersects with internal contour **502** of the region, the spiral tool path segment **514** is terminated at the point of intersection, possibly creating one or more separate unmachined residual areas generally adjacent to internal contour **502**. For each such separate residual area, if the size of the separate residual area is larger than a predetermined small value, it is calculated to be machined by a trochoidal-like tool path segment.

[0157] In a case where new linear subsegment **520** intersects with an island, the calculation of spiral tool path segment **514** is terminated at the point of intersection, and a moat is calculated to commence at the point of intersection and circumvent the island. The remainder of the region for which a tool path has yet to be calculated is designated as a new Type I region to be calculated separately.

[0158] The term “moat” is used throughout to designate a trochoidal-like tool path segment which machines a channel generally adjacent to an island that circumvents the island, thereby separating the island from the remainder of the material which needs to be machined. The width of the moat is preferably at least 2.5 times the radius of the cutting tool and preferably at most 4 times the radius of the cutting tool. These values are predefined, however they may be modified by the tool path designer. It is a particular feature of the present invention that machining a moat around an island is operative to create a residual region which is of the same type as the original region. This is of particular value when machining a Type I region or a Type III region which are thus able to be generally machined by spiral tool path segments which are generally more efficient than trochoidal-like tool path segments.

[0159] Additionally, the machining of a moat to circumvent an island is effective in preventing the formation of two fronts of a machined region adjacent to the island, which may potentially form one or more long narrow residual walls between the two fronts. As known to persons skilled in the art, the formation of narrow residual walls is undesirable as machining them may lead to damage to the cutting tool and\or to the workpiece.

[0160] Once second spiral point **518** has been calculated, the remaining number of loops to be included in the remainder of tool path segment **514** is updated. It is appreciated that the remaining number of loops may be a mixed number. The subsequent segments of the remainder of spiral tool path segment **514** are calculated recursively, whereby second spiral point **518** is designated to be a new first point of the remainder of spiral tool path segment **514**, and the bridge **530** immediately adjacent to second bridge **516** in a climbing direction of the cutting tool from second bridge **516** is designated to be a new second bridge. Additionally, second spiral point **518** is designated as a new bridge point of second bridge **516**, and the remaining region to be machined is recalculated.

[0161] The machining of a Type II region is calculated as follows:

[0162] Initially, a spiral machining time is calculated as the sum of the machining time needed for machining separating channels adjacent to all closed edges of the Type II region and the machining time needed for machining the remaining area of the region by a converging spiral tool path segment. Additionally, a trochoidal-like machining time is calculated as the machining time needed for machining the entire Type II region by a trochoidal-like tool path segment. If the spiral machining time is shorter than the trochoidal-like machining time, separating channels are calculated adjacent to all closed edges of the region, and the remaining separated area is calculated to be machined by a converging spiral tool path segment. If the spiral machining time is longer than the trochoidal-like machining time, a trochoidal-like tool path segment is calculated as follows:

[0163] The longest open edge of the region is selected as the “front” of the region. The remainder of the exterior boundary of the region is defined as the “blocking boundary”. A starting end is selected as one of the two ends of the front, for which when machining along the front from the starting end to the opposite end would result in a climb milling tool path.

[0164] As shown in FIG. **4**B a plurality of bridge lines **550** of a predefined density are each stretched from a front **552** across the region towards a blocking boundary **554**. A bridge point **556** of each of bridges **550** is initially defined as the point of intersection of each of bridges **550** with front **552**. A starting end **560** and an opposite end **562** are selected so that bridges **550** are ordered from starting end **560** to opposite end **562** in a climbing direction of the cutting tool. A single open trochoidal-like tool path segment **564** for machining an area adjacent to front **552** having a width which is generally equal to the maximum stepover is calculated by selecting a suitable point on each of bridges **550** and interconnecting the suitable points in the order of bridge lines **550** between starting end **560** and opposite end **562**, as follows:

[0165] Initially, starting end **560** is preferably selected as a first point of the single trochoidal-like tool path segment **564**. A possible second point of the trochoidal-like tool path segment **564** is calculated as a point on a first bridge **554**, immediately adjacent to first point **560** in a climbing direction of the cutting tool from first point **560**, which possible second point is distanced from bridge point **572** of the first bridge by the larger of the maximum stepover and the length of first bridge **554**. In the illustrated example of FIG. **4**B, the possible second point is calculated to be at the intersection **574** of first bridge **572** and blocking boundary **554**.

[0166] For the possible second point, the engagement angle at which the cutting tool will engage the material by following the cutting tool path from the first point to the possible second point is calculated. In a case where the calculated engagement angle is between the predetermined minimum and maximum engagement angles, the possible second point is chosen as the second point, and a new linear subsegment between first point **560** and the second point is added to the single trochoidal-like cutting tool path segment **564**.

[0167] In a case where the engagement angle is less than the predetermined minimum engagement angle, a binary search for a second point for which the calculated engagement angle is generally equal to the predetermined minimum engagement angle is performed. The binary search is performed between the possible second point and a point on first bridge **554** distanced from bridge point **572** of first bridge **554**, along first bridge **554**, by the larger of the maximum stepover and the length of first bridge **554**. Once a second point is found, a new linear subsegment between first point **560** and the second point is added to the single trochoidal-like cutting tool path segment **564**.

[0168] In a case where the engagement angle is greater than the predetermined maximum engagement angle, a binary search for a second point for which the calculated engagement angle is generally equal to the predetermined maximum engagement angle is performed. The binary search is performed between bridge point **572** of first bridge **554** and the possible second point. Once a second point is found, a new linear subsegment between first point **560** and the second point is added to the single trochoidal-like cutting tool path segment **564**.

[0169] In the illustrated example of FIG. **4**B, intersection **574** is selected as the second point, and a new linear subsegment **580** between first point **560** and second point **574** is added to the single trochoidal-like cutting tool path segment **564**.

[0170] Subsequently, calculation of the remainder of the single trochoidal-like tool path segment **564** is achieved by recursively performing the aforementioned calculation of tool path subsegments through suitable points on ordered bridges **550** up until opposite end **562** of selected front **552**. In a case where the single trochoidal-like tool path segment **564** crosses an island, the single trochoidal-like tool path segment **564** is clipped at the intersecting points of the single trochoidal-like tool path segment **564** and the external boundary of the island, thereby creating two disjoint subsegments of the single trochoidal-like tool path segment **564**. These two subsegments are then connected along a section of the external boundary of the island facing the front, which section is a closed edge.

[0171] The aforementioned calculation completes the calculation of a tool path segment for machining an object of the Type II region. At this point, the remainder of the Type II region to be machined is calculated, and a tool path for machining of the remainder of the Type II region is calculated recursively as described hereinabove. It is appreciated that the machining of the remainder of the Type II region requires repositioning of the cutting tool to a starting end of a front of the remainder of the Type II region. It is appreciated that repositioning techniques are well known to persons skilled in the art.

[0172] Referring now to the calculation of a tool path for machining of a Type III region, a diverging spiral tool path is preferred when machining Type III regions, as described hereinabove. A diverging spiral tool path segment calculated to machine a Type III region is a tool path segment which spirals outwardly from an innermost contour to an external boundary via a multiplicity of nested internal contours. The nested internal contours are calculated as follows:

[0173] A first nested internal contour is calculated to be a contour which is offset interiorly to the external boundary of the region by a distance which is generally equal to 1.5 radii of the cutting tool. Additional nested internal contours are then calculated recursively inwardly from the first nested internal contour, each nested internal contour being inwardly spaced from the nested internal contour immediately externally adjacent thereto by a distance which is generally equal to 1.5 radii of the rotating cutting tool. A last nested internal contour is calculated to be a contour having a center of area which is closer than 1.5 radii of the cutting tool to at least one point on the contour. Inwardly of the last nested internal contour, the innermost contour is calculated to be a small circle having a radius which is generally smaller than the radius of the cutting tool, and which is centered around the center of area of the last nested internal offset contour.

[0174] In a case where the innermost contour is either within the external boundary of an island or intersects with the external boundary of an island, a moat is calculated to circumvent the island, and the innermost contour is recalculated to be immediately external to the external boundary of the moat, such that the innermost contour does not intersect with any other islands. It is noted that nested internal contours which intersect with an external boundary of any island are discarded.

[0175] Once the nested internal contours have been calculated, the number of loops to be included in a diverging spiral tool path segment which will spiral outwardly from the innermost contour to the last nested internal offset contour is calculated preferably as follows:

[0176] A plurality of bridge lines are stretched from the innermost contour to a next internal offset contour immediately externally adjacent thereto. A bridge point of each bridge is initially defined as the point of intersection of the bridge with the innermost contour. The length of the shortest bridge divided by the minimum stepover provides a theoretical maximum of the number of loops that can be theoretically included in the diverging spiral tool path. The length of the longest bridge divided by the maximum stepover provides an absolute minimum of the number of loops which must be included in the diverging spiral tool path segment that is required to machine the entire area between the innermost contour and the next internal offset contour.

[0177] It is appreciated that the furthest distance, in any direction, from the innermost contour which can be reached by a diverging spiral tool path segment is the number of loops included in the diverging spiral tool path segment multiplied by the maximum stepover. Areas between the innermost contour and the next internal offset contour beyond this furthest distance cannot be machined by the diverging spiral tool path segment, and are preferably machined by clipping after the machining of the diverging spiral tool path segment.

[0178] Throughout, the parameter n is used to designate a possible number of loops to be included in the spiral tool path segment, wherein n is a number between the minimum number of loops which must be included in the spiral tool path segment and the maximum number of loops that can be included in the spiral tool path segment.

[0179] For each possible value of n, the work time needed to machine the area between the innermost contour and the next internal offset contour is calculated by summing the time needed to machine the spiral tool path segment and the time needed to machine all clipped areas which were identified between the innermost contour and the next internal offset contour as described hereinabove. The optimal value of loops to be included in the spiral tool path segment is chosen to be the value of n for which the calculated work time is the shortest.

[0180] Once the optimal value of loops to be included in the tool path segment is chosen, the actual path of the spiral tool path segment is calculated. Initially, the bridge point of a first bridge is preferably selected as a starting spiral point of the spiral tool path segment. The first bridge is preferably selected to minimize the time required to move the rotating cutting tool from its previous position. A possible second spiral point of the spiral tool path segment is calculated as a point on a second bridge, immediately adjacent to the first bridge in a climbing direction of the cutting tool from the first bridge, which point is distanced from the bridge point of the second bridge by the length of the second bridge divided by the remaining number of loops to be included in the tool path segment.

[0181] For the possible second spiral point, the engagement angle at which the cutting tool will engage the material by following the cutting tool path from the first spiral point to the possible second spiral point is calculated. In a case where the calculated engagement angle is between the predetermined minimum and maximum engagement angles, the possible second spiral point is chosen as the second spiral point, and a new linear subsegment between the first spiral point and the second spiral point is added to the spiral cutting tool path segment.

[0182] In a case where the engagement angle is less than the predetermined minimum engagement angle, a binary search for a second spiral point for which the calculated engagement angle is generally equal to the predetermined minimum engagement angle is performed. The binary search is performed between the possible second spiral point and a point on the second bridge distanced from the bridge point of the second bridge by the maximum stepover. Once a second spiral point is found, a new linear subsegment between the first spiral point and the second spiral point is added to the spiral tool path segment.

[0183] In a case where the engagement angle is greater than the predetermined maximum engagement angle, a binary search for a second spiral point for which the calculated engagement angle is generally equal to the predetermined maximum engagement angle is performed. The binary search is performed between the bridge point of the second bridge and the possible second spiral point. Once a second spiral point is found, a new linear subsegment between the first spiral point and the second spiral point is added to the spiral tool path segment.

[0184] In a case where the new linear subsegment intersects with an island, the calculation of the spiral tool path segment is terminated at the point of intersection, where a moat is calculated to commence and circumvent the island. The remainder of the region for which a tool path has yet to be calculated is designated as a new Type III region to be calculated separately.

[0185] In a case where the new linear subsegment intersects with the next internal offset contour, an additional loop of the diverging spiral tool path segment is calculated, and the portions of the additional loop which are internal to the next internal offset contour define one or more uncalculated residual regions between the diverging spiral tool path segment and the next internal offset contour, which residual regions are each calculated as a Type II region, preferably by employing a trochoidal-like tool path segment. The portions of the additional loop which are internal to the next internal offset contour are connected along the next internal offset contour to form a continuous loop which is the final loop of the diverging spiral tool path segment.

[0186] Once the second spiral point has been calculated, the remaining number of loops to be included in the tool path segment is recalculated and the subsequent segments of the spiral cutting tool path segment are calculated recursively, whereby the second spiral point is designated to be a new starting point of the remainder of the spiral tool path segment, and the bridge immediately adjacent to the second bridge in a climbing direction of the cutting tool from the second bridge is designated to be the new second bridge. Additionally, the second spiral point is designated as the new bridge point of the second bridge, and the remaining region to be machined is recalculated.

[0187] Subsequently, calculation of the remainder of the diverging spiral tool path for the remainder of the region is achieved by recursively performing the aforementioned calculation of diverging spiral tool path segments through subsequent consecutive pairs of nested internal contours between the last nested internal offset contour and the external boundary of the region.

[0188] It is appreciated that all of the calculations of the tool paths described hereinabove produce piecewise linear tool paths. In cases where a piecewise linear tool path is not suitable for a particular workpiece being machined by a particular CNC machine, a smoothing approximation of the piecewise linear tool path may be calculated. Such approximation methods are well known to persons skilled in the art.

[0189] [**0**.] Returning now to the illustrated example of FIG. **1**D, cross section **116** is initially identified as a Type I region which includes multiple protrusions. Therefore, a converging spiral tool path segment is calculated between the external boundary of the workpiece and a calculated internal contour, as the initial tool path segment. This calculation preferably begins with calculation of a spiral tool path segment which begins from a selected location just outside the periphery of cross section **116**. Reference is made in this context to FIGS. **1**E-**1** and **1**E-**2**, which are respective isometric and top view illustrations of the stock material **114** overlaid by outline **121** of object **100** in which the initial spiral tool path segment is indicated generally by reference numeral **122**. It is noted that the spiral tool path is indicated by solid lines, which represent the center of the rotating cutting tool, whose cross-sectional extent is designated by reference numeral **124** in FIG. **1**E-**2**. The selected location, here designated by reference numeral **126**, is preferably selected to minimize the

time required to move the rotating cutting tool from its previous position.
[0190] In the illustrated example of FIGS. **1**E-**1** and **1**E-**2**, the initial tool path segment is a converging spiral segment which is calculated as described hereinabove. As shown in FIGS. **1**E-**1** and **1**E-**2**, initial spiral tool path segment **122**, ultimately intersects with island **105** at intersecting point **130** at which point spiral tool path segment **122** is terminated. As shown in FIGS. **1**F-**1** and **1**F-**2**, a moat **132** which circumvents island **105** is calculated.
[0191] As shown in FIGS. **1**F-**1** and **1**F-**2**, an inner boundary **134** of moat **132** is calculated to be generally alongside the outer boundary of island **105**. It is appreciated that a narrow offset remains between the island **105** and an inner boundary **134** of the moat, which may be finish machined at a later stage. The outer boundary **136** of moat **132** is calculated as being offset from inner boundary **134** by the moat width.
[0192] As shown in FIGS. **1**F-**1** and **1**F-**2**, the outer boundary **136** of moat **132** intersects with island **107** at points **138** and **139**. Therefore, an additional moat **140** is calculated to circumvent island **107**, whereby moats **132** and **140** are joined to form one continuous moat which circumvents islands **105** and **107**. As clearly shown in FIGS. **1**F-**1** and **1**F-**2**, the combination of the initial spiral tool path segment **122** and subsequent moats **132** and **140** which circumvent islands **105** and **107** define a new Type I region which is designated by reference numeral **142**.
[0193] Region **142** includes multiple islands **109**, **111** and **113**. As clearly shown in FIGS. **1**F-**1** and **1**F-**2**, a bottleneck **150** is detected in region **142**. Therefore, as shown in FIGS. **1**G-**1** and **1**G-**2**, a separating channel **152** is calculated at the location of bottleneck **150**, effectively dividing region **142** into two independent Type I regions designated by reference numerals **154** and **156**.
[0194] Turning now to FIGS. **1**H-**1** and **1**H-**2**, it is shown that initially, a spiral tool path segment for region **154** is calculated, while the calculation of region **156** is deferred. As shown in FIGS. **1**H-**1** and **1**H-**2**, a starting point **160** is chosen and a spiral tool path segment **162** extends from initial point **160** generally along the external boundary of region **154** until intersecting with island **109** at intersecting point **164** at which point spiral tool path segment **162** is terminated. As shown in FIGS. **1**I-**1** and **1**I-**2**, a moat **166** which circumvents island **109** is calculated. The remainder of region **154** is identified as a Type I region designated by reference numeral **170**.
[0195] Region **170** includes islands **111** and **113**. As clearly shown in **1**I-**1** and **1**I-**2**, a bottleneck **172** is detected in region **170**. Therefore, as shown in **1**J-**1** and **1**J-**2**, a separating channel **174** is calculated at the location of bottleneck **172**, effectively dividing region **170** into two independent Type I regions designated by reference numerals **176** and **178**.
[0196] Turning now to **1**K-**1** and **1**K-**2**, it is shown that initially, a spiral path for machining region **176** is calculated, while the calculation of region **178** is deferred. As shown in FIGS. **1**K-**1** and **1**K-**2**, region **176** does not include any islands, therefore a converging spiral tool path segment is calculated to machine region **176** with the internal boundary of region **176** being a small circle **177** of a radius which is generally smaller than the radius of the tool, and which is centered around the center of area of region **176**.
[0197] Subsequentially, a spiral tool path segment for region **178** is calculated. As shown in FIGS. **1**L-**1** and **1**L-**2**, a starting point **180** is chosen and a spiral tool path segment **182** is extended from initial point **180** generally along the external boundary of region **178** until intersecting with island **111** at intersecting point **184** at which point spiral tool path segment **182** is terminated. As shown in FIGS. **1**M-**1** and **1**M-**2**, a moat **186** which circumvents protrusion **110** is calculated.
[0198] It is appreciated that in a case where the external boundary of a moat is calculated to be in close proximity to the external boundary of the Type I region which includes the moat, a local widening of the moat is calculated to prevent the forming of a narrow residual wall between the moat and the external boundary of the region. As known to persons skilled in the art, the formation of narrow residual walls is undesirable as machining them may lead to damage to the cutting tool and\or to the workpiece.
[0199] As seen in FIGS. **1**M-**1** and **1**M-**2**, the external boundary of moat **186** is calculated to be in

close proximity to the external boundary of region **178**. Therefore, moat **186** is locally widened up to the external boundary of region **178**, along a narrow residual wall area **189** in which, without this widening, a narrow residual wall would be have been formed between moat **186** and the external boundary of region **178**. Locally widened moat **186** divides region **178** into two independent Type I regions designated by reference numerals **190** and **192**.

[0200] Turning now to Figs. IN-**1** and **1**N-**2**, it is shown that initially, region **190** is calculated, while the calculation of region **192** is deferred. As shown in Figs. **1**N-**1** and **1**N-**2**, two clipped areas of region **190** designated by numerals **196** and **198** are identified. Areas **196** and **198** are calculated to be machined by a trochoidal-like tool path segment prior to the machining of the remainder of region **190** by a spiral tool path segment.

[0201] The remainder of region **190** does not include any islands, therefore a converging spiral tool path segment is calculated to machine the remainder of region **190** with the internal boundary being a small circle **191** of a radius which is generally smaller than the radius of the tool, and which is centered around the center of area of the remainder of region **190**.

[0202] Subsequentially, a spiral tool path segment for region **192** is calculated. As shown in FIGS. **1**O-**1** and **1**O-**2**, one area of region **192**, designated by numeral **200** is identified by clipping. Area **200** is calculated to be machined by a trochoidal-like tool path segment prior to the machining of the remainder of region **192** by a spiral tool path segment.

[0203] Additionally, as shown in FIGS. **1**P-**1** and **1**P-**2**, an additional area of region **192**, designated by numeral **202** is identified by clipping. However, it is calculated that area **202** would be more efficiently machined as a separate Type I region. Therefore, a separating channel **210** which divides the remainder of region **192** into two Type I regions designated by numerals **202** and **214** is calculated. Region **202** does not include any protrusions, therefore, as shown in FIGS. **1**Q-**1** and **1**Q-**2**, a converging spiral tool path segment is calculated to machine region **202** with the internal boundary being a small circle **213** of a radius which is generally smaller than the radius of the tool, and which is centered around the center of area of region **202**.

[0204] It is calculated that machining a separating channel **210** and machining region **202** as a Type I region results in a machining time which is shorter than the machining time of region **202** by a trochoidal-like tool path segment.

[0205] Turning now to FIGS. **1**R-**1** and **1**R-**2**, it is shown that region **214** includes one island **113** which is generally centrally located within region **214**. Therefore, a converging spiral tool path segment **216** is calculated to machine region **214** with the internal boundary being generally alongside the external perimeter of island **113**. As shown in FIG. **1**R-**2**, spiral tool path segment **216**, ultimately intersects with island **113** at intersecting point **218** at which point spiral tool path segment **216** is terminated. It is appreciated that after machining segment **216**, there may remain one or more Type II regions adjacent to island **113** which are machined by trochoidal-like tool path segments.

[0206] Turning now to FIGS. **1**S-**1** and **1**S-**2**, it is shown that, the machining of region **156** is calculated. As shown in FIGS. **1**S-**1** and **1**S-**2**, a clipped area of region **156** designated by numeral **230** is identified by clipping. Area **230** is preferably calculated to be machined by a trochoidal-like tool path segment, and the remainder of region **156** is then calculated to be machined by a spiral tool path segment.

[0207] It is appreciated that the calculation described hereinabove constitutes the calculation of a tool path for the machining of a first step down which is a first phase in the machining of object **100**. Throughout, the term “step down” is used to describe a single machining phase at a constant depth. As shown in FIG. **1**C, the complete machining of object **100** requires three step downs. Therefore, subsequent and similar to the calculation described hereinabove, the tool path designer calculates the machining of second step down **119** and thereafter of third step down **120**, thereby completing the entire rough machining of object **100**. Preferably, the vertical distance between subsequent step downs is generally between 1 and 4 times the diameter of the cutting tool.

[0208] It is appreciated that following the rough machining of a workpiece, an additional stage of rest rough machining is calculated, which reduces the large residual steps created by the series of step downs on the sloping surfaces of object **100**.

[0209] Reference is now made to FIGS. **2**A-**2**L-**2**, which illustrate the calculation of another tool path in accordance with a preferred embodiment of the present invention. FIGS. **2**A and **2**B are respective isometric and top view illustrations of an object **400**, which is another example of objects that can be fabricated in accordance with the present invention. The configuration of the object **400** is selected to illustrate additional various particular features of the present invention. It is noted that any suitable three-dimensional object that can be machined by a conventional 3-axis machine tool may be fabricated in accordance with a preferred embodiment of the present invention.

[0210] As seen in FIGS. **2**A & **2**B, the object **400** is seen to have a generally planar base portion **402** from which one protrusion, here designated by reference numeral **404**, extends. FIG. **2**C shows stock material **410** overlaid by a cross section **420** of object **400**. Cross section **420** is characterized as having an external boundary **422** and an island **405** corresponding to the cross section of protrusion **404** at the depth of cross section **420**.

[0211] In the illustrated example of FIG. **2**C, cross section **420** is initially identified as a Type III region **424** which includes one island **405**. As described hereinabove, a plurality of nested offset contours is calculated between the external boundary **422** of region **424** and an innermost contour of region **424**. The innermost contour is initially calculated to be overlapping with the external boundary of island **405**. Therefore, as shown in FIGS. **2**D-**1** and **2**D-**2**, a moat **428** is calculated to circumvent island **405**, and the innermost contour **430** is calculated to be immediately external to the external boundary of moat **428**.

[0212] As shown in FIGS. **2**D-**1** and **2**D-**2**, innermost contour **430** and nested internal contour **440**, external to innermost contour **430**, define a Type III region **442**. As shown in FIGS. **2**E-**1** and **2**E-**2**, a diverging tool path segment **443** is initially calculated to spiral outwardly between innermost contour **430** and nested internal contour **440**, thereby creating two residual regions **444** and **446**. As shown in FIGS. **2**F-**1** and **2**F-**2**, residual region **444** is calculated to be machined as a Type II region by employing a trochoidal-like tool path segment. Similarly, as shown in FIGS. **2**G-**1** and **2**G-**2**, residual region **446** is calculated to be machined as a Type II region by employing a trochoidal-like tool path segment.

[0213] Turning now to FIGS. **2**H-**1** and **2**H-**2**, it is shown that a diverging spiral tool path segment is calculated to machine a Type III region **448** defined between nested internal contour **440** and nested internal contour **450**. Subsequently, as shown in FIGS. **2**I-**1** and **2**I-**2**, a diverging spiral tool path segment is similarly calculated to machine a Type III region **452** defined between nested internal contour **450** and nested internal contour **460**.

[0214] Turning now to FIGS. **2**J-**1** and **2**J-**2**, it is shown that a diverging tool path segment is calculated to machine a Type III region **468** defined between nested internal contour **460** and external boundary **422**, thereby creating two residual regions **470** and **472**. As shown in FIGS. **2**K-**1** and **2**K-**2**, residual region **470** is calculated to be machined as a Type II region by employing a trochoidal-like tool path segment. Similarly, as shown in FIGS. **2**L-**1** and **2**L-**2**, residual region **472** is calculated to be machined as a Type II region by employing a trochoidal-like tool path segment, thereby completing the calculation of the machining of object **400**.

[0215] Reference is now made to FIG. **5**, which is a simplified composite pictorial and sectional illustration of an initial workpiece **600**, a machined object **602** formed from the workpiece in accordance with a preferred embodiment of the present invention and a finished object **604** to be produced from the machined object. In the illustrated embodiment, the initial workpiece **600** is shown as a block of metal such as, for example, tool steel, mold steel or titanium. The machined object **602** is seen as a roughed out object which will be finished using techniques, which are outside the scope of the present invention, to produce the finished object **604**.

[0216] For simplicity of explanation, a common vertical axis Z is defined in the workpiece **600**, in the machined object **602** and in the finished object **604**.
[0217] The machined object **602** of FIG. **5** is typically characterized in that it is a generally disc-like object, which is typically non-circularly symmetric and typically has non-uniformly sloped edges.
[0218] More specifically referring to the illustrated machined object **602** and finished object **604**:
[0219] the slope configuration of the edge at least a given height along the Z axis varies at different azimuthal locations along the edge; and [0220] the slope of the edge varies at different heights along the Z axis at at least one given azimuth.
[0221] It is appreciated that machined objects may be produced in accordance with an embodiment of the present invention wherein only one or neither of these features exist. For the purposes of explanation, the shapes of the machined object **602** and of the finished object **604** have been selected to illustrate both of these features.
[0222] The method of the present invention described below with reference to FIGS. **6-12**C is an automated computer-implemented system and method for generating commands for controlling a computer numerically controlled milling machine to fabricate a machined object from a workpiece having a Z-axis, the machined object being configured to facilitate subsequent finishing into a finished object. An illustrated embodiment of the method includes the following steps: [0223] defining a surface of the finished object; [0224] defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object; [0225] defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object; and [0226] calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein: [0227] surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface; and [0228] the number of multiple step-up cuts in the workpiece at multiple heights along the Z-axis and the areas cut in each of the multiple step-up cuts are selected so as to generally minimize the amount of workpiece material that is removed from the workpiece during the cuts while ensuring that the surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface.
[0229] In another aspect, the system and method provide the following: [0230] defining a surface of the finished object; [0231] defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting surface of the machined object; [0232] defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object; and [0233] calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein: [0234] surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface; and [0235] a decision of whether or not to cut the workpiece at a given location at each height of each of the multiple step-up cuts is a function of the required non-vertical slope of the finished object at the given location.
[0236] In a further aspect, the method and system provide: [0237] defining a surface of the finished object; [0238] defining an offset surface, the offset surface being outside the surface of the finished object and separated therefrom by an offset distance, the offset surface defining an inner limiting

surface of the machined object; [0239] defining a scallop surface, the scallop surface being outside the offset surface and separated therefrom by a scallop distance, the scallop surface defining an outer limiting surface of the machined object; and [0240] calculating a tool path for the computer numerically controlled milling machine which produces multiple step-up cuts in the workpiece at multiple heights along the Z-axis, the multiple step up cuts in the workpiece resulting in the machined object, wherein: [0241] surfaces of the machined object produced by the multiple step-up cuts all lie between the inner limiting surface defined by the offset surface and the outer limiting surface defined by the scallop surface; and [0242] a decision as to at which height each of the multiple step-up cuts is made is a function of the required non-vertical slope of the finished object at the given height at various locations on the finished object.

[0243] Reference is now made to FIG. **6**, which is a simplified annotated pictorial illustration of the machined object **602** of FIG. **5** showing, in sectional enlargements here designated by letters A and B, typical finished object surface, offset surface and scallop surface profiles at two mutually azimuthally separated locations, respectively designated A and B, in accordance with a preferred embodiment of the present invention.

[0244] FIG. **6** is annotated to show three mutually parallel curves: [0245] a first curve, here designated as finished object surface curve **610**, which represents the intended wall surface of a finished object to be produced from machined object **602**, by finishing techniques which are outside the scope of the present invention; [0246] a second curve, here designated as surface offset curve **620**, parallel to finished object surface curve **610** in three dimensions, which represents an offset distance from the finished object surface curve **610**, indicating a minimum thickness of material which must remain on the machined object **602** beyond the finished object surface curve **610**; [0247] a third curve, here designated as scallop surface curve **630**, parallel to finished object curve **610** and to surface offset curve **620** in three dimensions, which represents a maximum scallop distance from the surface offset curve **620** defining the maximum thickness of metal which can remain on the machined object **602** beyond the surface offset curve.

[0248] In a typical case, the separation between the finished object surface curve **610** and the surface offset curve **620** is a few millimeters and the separation between the surface offset curve **620** and the scallop surface curve **630** is 10% to 50% of the separation between the finished object surface curve **610** and the surface offset curve **620**. The scallop tolerance is typically 10% to 30% of the separation between the surface offset curve **620** and the scallop surface curve **630**.

[0249] The finished object surface curve **610** is typically selected by the object designer. The separations between the finished object surface curve **610**, the surface offset curve **620** and the scallop surface curve **630**, as well as a scallop tolerance are typically selected by a computerized tool path technologist who employs an embodiment of the present invention for programming tool paths for computerized machine tools.

[0250] Criteria used by the computerized tool path technologist in selecting the foregoing separations between the finished object surface curve **610**, the surface offset curve **620** and the scallop surface curve **630** as well as the scallop tolerance are well known.

[0251] Reference is now made additionally to FIG. **7**, which is a simplified pictorial illustration of the initial workpiece **600** of FIG. **5** showing an initial deep cut **650** therein and also showing, in enlargements respectively designated A and B, the corresponding enlargements A and B of FIG. **6** superimposed over the workpiece **600**, having the initial deep cut **650**. In the illustrated embodiment, the initial deep cut **650** is circumferential, it being appreciated that this is not necessarily the case.

[0252] The initial circumferential deep cut **650** defines a deep cut floor surface **652**, a circumferential deep cut edge wall surface **654** and an intersection curve **656** which represents the intersection between deep cut floor surface **652** and circumferential deep cut edge wall surface **654**.

[0253] In accordance with a preferred embodiment of the present invention, once the surface offset curve **620** and the scallop surface curve **630** have been established, a tool path is generated for the initial deep cut **650** shown in FIG. **7**. This tool path is shown in FIG. **8** and designated by reference numeral **660**. The outside edge of the workpiece **600** is indicated by reference numeral **680**.

[0254] Thereafter, tool paths are generated for cutting steps into the edge wall **654** of the workpiece, preferably sequentially and monotonically upward from the deep cut floor surface **652** defined by initial deep cut **650**, in accordance with a preferred embodiment of the present invention. These monotonically upwardly cut steps are here termed “step-up cuts” and transform the workpiece shown in FIG. **7** into the machined object **602** of FIG. **5**.

[0255] Reference is now made in this context additionally to FIGS. **9**A, **9**B and **9**C, which are simplified sectional illustrations inter alia of the corresponding annotated sectional illustrations in the respective enlargements A and B of FIG. **6** superimposed over the workpiece **600** having the deep cut **650**, as it appears in FIGS. **7** and **8** and respectively showing first, second and third step-up cuts in accordance with a preferred embodiment of the present invention.

[0256] A preferred method for calculation of the height of the first step-up cut will now be described in detail with reference to FIG. **9**A:

[0257] Initially, the height of a first step from the bottom of the initial deep cut is calculated for a collection of mutually azimuthally separated points **682** densely distributed all along curve **656** at the intersection of floor **652** and the edge wall **654**, defined by the initial deep cut **650** shown in FIGS. **7** and **8** as follows:

[0258] For each one of the collection of points **682**, an imaginary vertical line **684**, parallel to the Z-axis and along the edge wall **654**, is constructed to extend through the point **682** and intersect at a point **686** with the scallop surface curve **630**. The height of intersection point **686** is noted; and

[0259] Taking the heights of the intersection points **686** of the imaginary vertical lines **684** with the scallop surface curve **630** for each point **682** in the collection of points all along the circumference of edge wall **654**, the lowest height of an intersection point from among all of them is selected as the first step height and is identified as a point along line **684** and designated by reference numeral **688**. A curve running through all points **688** along the circumference of edge wall **654** is designated by reference numeral **689**.

[0260] Next, a determination of which contiguous azimuthal regions along the circumference of the initial deep cut, represented by edge wall **654**, are to be cut down to the height represented by point **688**, is carried out as follows:

[0261] An imaginary slice is taken through the workpiece of FIGS. **7**, **8**, **9**A, **9**B & **9**C perpendicular to common vertical axis Z at the first step height represented by point **688** and is shown as a horizontal line designated **690** in FIGS. **9**A, **9**B & **9**C;

[0262] The normal distance **692** between the point **688** at height **690** and the scallop surface curve **630** is ascertained;

[0263] If distance **692** is less than the scallop tolerance, which is a predetermined value, the point **682** is marked as a “good to cut” point, here designated by the letter Y in FIG. **10**A. Otherwise that point **682** is marked as a “no cut” point, here designated by the letter N in FIG. **10**A.

[0264] Once all of the points **682**, which typically number in the tens of thousands in large machined objects, have been classified as either Y points or N points, as seen in FIG. **10**A, a short “no-cut” gap elimination process is undertaken wherein sequences of “no-cut” points **682** of less than a predetermined number of points, typically 4 points, are reclassified as “good to cut” points. FIG. **10**A indicates such short “no-cut” gaps and their reclassification and FIG. **10**B shows the reclassified sequences of points.

[0265] For each sequence of at least 4 contiguous “good to cut” points, an azimuthal cut region **700** is defined as illustrated in FIG. **10**C and for each sequence of at least 4 contiguous “no-cut” points, an azimuthal non-cut region **702** is defined as illustrated in FIG. **10**C.

[0266] It is appreciated that the depth of cut in a plane, perpendicular to the Z-axis, at a height represented by horizontal line **690**, which indicates the width of the cut region **700** corresponding to each “good to cut” point **682** in FIG. **10**B is determined by the separation between curve **689** and the intersection of the plane defined by horizontal line **690** with the offset surface **620**. The intersection of the plane defined by horizontal line **690** with the offset surface **620** is a curve designated by reference numeral **720** and defines the base of a second step forward edge wall **722** and an edge of a first step floor surface **724** which lies at height **690**.

[0267] It is further appreciated that the cut region **700** is preferably machined as a semi-open region as described hereinabove with reference to FIG. **4**B.

[0268] Calculation of the height of a second step-up cut will now be described in detail with reference to FIG. **9**B:

[0269] Initially, the height of a step from first step floor surface **724**, which height represents the height of the second step, is calculated for a collection of mutually azimuthally separated points **732** densely distributed all along curve **720** representing the intersection of second step edge wall **722** with first step floor surface **724**, as shown in FIGS. **9**B & **9**C as follows:

[0270] For each one of the collection of points **732**, an imaginary vertical line **734**, parallel to the Z-axis and along the second step edge wall **722**, is constructed to extend through the point **732** and intersect at a point **736** with the scallop surface curve **630**. The height of intersection point **736** is noted;

[0271] Taking the heights of the intersection points **736** of the imaginary vertical lines **734** with the scallop surface curve **630** for each point **732** in the collection of points all along the circumference of second step edge wall **722**, the lowest height of an intersection point from among all of them is selected as the second step height and is identified as a point along line **734** and designated by reference numeral **738**. A curve running through all points **738** along the circumference of second step edge wall **722** is designated by reference numeral **739**.

[0272] An imaginary horizontal slice is taken through the workpiece of FIGS. **7**, **8**, **9**A, **9**B & **9**C at the second step height represented by point **738** and is shown as a horizontal line designated **740** in FIGS. **9**B & **9**C;

[0273] The normal distance **741** between the point **738** at height **740** and the scallop surface curve **630** is ascertained;

[0274] If normal distance **741** is less than the scallop tolerance, which is a predetermined value, the point **732** is marked as a “good to cut” point, here designated by the letter Y in FIG. **11**A. Otherwise that point **732** is marked as a “no cut” point, here designated by the letter N in FIG. **11**A.

[0275] Once all of the points **732**, which typically number in the tens of thousands in large machined objects, have been classified as either Y points or N points, as seen in FIG. **11**A, a short “no-cut” gap elimination process is undertaken wherein sequences of “no-cut” points **732** of less than a predetermined number of points, typically 4 points, are reclassified as “good to cut” points. FIG. **11**A indicates such short “no-cut” gaps and their reclassification and FIG. **11**B shows the reclassified sequences of points.

[0276] For each sequence of at least 4 contiguous “good to cut” points, an azimuthal cut region **742** is defined as illustrated in FIG. **11**C and for each sequence of at least 4 contiguous “no-cut” points, an azimuthal no-cut region **744** is defined as illustrated in FIG. **11**C.

[0277] It is appreciated that the depth of cut in a plane, perpendicular to the Z-axis, at a height represented by horizontal line **740**, which indicates the width of the cut region **742** corresponding to each “good to cut” point **732** in FIG. **11**B is determined by the separation between curve **739** and the intersection of the plane defined by horizontal line **740** with the offset surface **620**. The intersection of the plane defined by horizontal line **740** with the offset surface **620** is a curve designated by reference numeral **750** and defines the base of a third step forward edge wall **752** and an edge of a second step floor surface **754** which lies at height **740**.

[0278] It is further appreciated that the cut region **742** is preferably machined as a semi-open region as described hereinabove with reference to FIG. **4**B.

[0279] Calculation of the height of a third step-up cut will now be described in detail with reference to FIG. **9**C:

[0280] Initially, the height of a step from second step floor surface **754**, which height represents the height of the third step, is calculated for a collection of mutually azimuthally separated points **762** densely distributed all along curve **750** representing the intersection of third step forward edge wall **752** with second step floor surface **754**, as shown in FIGS. **9**B & **9**C as follows:
[0281] For each one of the collection of points **762**, an imaginary vertical line **764**, parallel to the Z-axis and along the third step forward wall **752**, is constructed to extend through the point **762** and intersect at a point **766** with the scallop surface curve **630**. The height of intersection point **766** is noted; and
[0282] Taking the heights of the intersection points **766** of the imaginary vertical lines **764** with the scallop surface curve **630** for each point **762** in the collection of points all along the circumference of third step forward edge wall **752**, the lowest height of an intersection point from among all of them is selected as the third step height and is identified as a point along line **764** and designated by reference numeral **768**. A curve running through all points **768** along the circumference of third step forward edge wall **752** is designated by reference numeral **769**.
[0283] An imaginary horizontal slice is taken through the workpiece of FIGS. **7**, **8**, **9**A, **9**B & **9**C at the third step height represented by point **768** and is shown as a horizontal line designated **770** in FIG. **9**C;
[0284] The normal distance **771** between the point **768** at height **770** and the scallop surface curve **630** is ascertained;
[0285] If normal distance **771** is less than the scallop tolerance, which is a predetermined value, the point **762** is marked as a “good to cut” point, here designated by the letter Y in FIG. **12**A. Otherwise that point **762** is marked as a “no cut” point, here designated by the letter N in FIG. **12**A.
[0286] Once all of the points **762**, which typically number in the tens of thousands in large machined objects, have been classified as either Y points or N points, as seen in FIG. **12**A, a short “no-cut” gap elimination process is undertaken wherein sequences of “no-cut” points **762** of less than a predetermined number of points, typically 4 points, are reclassified as “good to cut” points. FIG. **12**A indicates such short “no-cut” gaps and their reclassification and FIG. **12**B shows the reclassified sequences of points.
[0287] For each sequence of at least 4 contiguous “good to cut” points, an azimuthal cut region **772** is defined as illustrated in FIG. **12**C and for each sequence of at least 4 contiguous “no-cut” points, an azimuthal no-cut region **774** is defined as illustrated in FIG. **12**C.
[0288] It is appreciated that the depth of cut in a plane, perpendicular to the Z-axis, at a height represented by horizontal line **770**, which indicates the width of the cut region **772** corresponding to each “good to cut” point **762** in FIG. **12**B is determined by the separation between curve **739** and the intersection of the plane defined by horizontal line **770** with the offset surface **620**. The intersection of the plane defined by horizontal line **770** with the offset surface **620** is a curve designated by reference numeral **780** and defines the base of a fourth step forward edge wall **782** and an edge of a third step floor surface **784** which lies at height **770**.
[0289] It is further appreciated that the cut region **772** is preferably machined as a semi-open region as described hereinabove with reference to FIG. **4**B.
[0290] The foregoing process continues until step-up cuts reaching the top of workpiece **600** have been calculated. Once all of the step-up cut calculations have been completed, the order in which the various semi-open cut regions are to be machined is determined by application of known techniques and methodologies, which are outside of the scope of the present invention.
[0291] It will be appreciated that the aforesaid method of calculation of the step-up cut regions has at least the following beneficial results:
[0292] Generally ensuring that machined object **602** does not include material which extends beyond the designated scallop surface; and
[0293] Generally ensuring that step-up machining of the workpiece **600** does not unnecessarily remove material which removal is not mandated by the designated scallop surface.

[0294] Reference is now made to FIG. **13**, which is a simplified partially symbolic, partially pictorial illustration of exemplary selection of axial depth of cut and stepover as a function of available spindle power in accordance with a preferred embodiment of the present invention.
[0295] The method of the present invention described below in reference to FIG. **13** is an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object from a workpiece. In the illustrated embodiment of the present invention, the method includes the following steps: [0296] ascertaining the available spindle power of the computer numerically controlled milling machine; [0297] automatically selecting, using a computer, a maximum depth and width of cut, which are a function at least of the available spindle power of the computer numerically controlled milling machine; and [0298] configuring a tool path for the tool relative to the workpiece in which the tool path includes a plurality of tool path layers whose maximum thickness and width of cut correspond to the maximum depth and width of cut.
[0299] FIG. **13** illustrates four typical milling machines having four different levels of available spindle power. For example, milling machine **800** is a Makino A99 milling machine with a 50 KW spindle; milling machine **810** is a Makino A88e milling machine with a 30 KW spindle; milling machine **820** is a Haas VF2 milling machine with a 15 KW spindle and milling machine **830** is a MUGA Center R45-30 with a 5.5 KW spindle.
[0300] For the purposes of explanation, it is assumed that an identical workpiece, typically a block of steel **850**, having typical dimensions of 300 mm by 300 mm by 150 mm, is machined by each of milling machines **800**, **810**, **820** and **830** to produce an identical machined object **860**. For clarity, the block **850** and the machined object **860** are shown out of proportion to the size of the milling machines.
[0301] As seen in FIG. **13**, milling machine **800**, which has a relatively high spindle power, removes material from block **850** in a single step and following a toolpath which has a maximum stepover, typically 3 mm. This is shown at machining stage **862** and the toolpath is shown schematically and designated by reference numeral **864**.
[0302] As further seen in FIG. **13**, milling machine **810**, which has a medium spindle power, may remove material from block **850** in one of two alternative procedures. In a first of such alternative procedures, milling machine **810** removes material from block **850** in a single step and following a toolpath which has an intermediate stepover, typically 1.8 mm. This is shown at machining stage **866** and the toolpath is shown schematically and designated by reference numeral **868**.
[0303] In a second of such alternative procedures, milling machine **810** removes material from block **850** in two steps and following a toolpath which has a maximum stepover, typically 3 mm. This is shown at machining stages **870** and **872** and the toolpaths are shown schematically and designated by reference numeral **874**.
[0304] As additionally seen in FIG. **13**, milling machine **820**, which has a low spindle power, removes material from block **850** in two steps and following a toolpath which has an intermediate stepover, typically 1.8 mm. This is shown at machining stages **876** and **877** and the toolpath is shown schematically and designated by reference numeral **878**.
[0305] As further seen in FIG. **13**, milling machine **830**, which has a very low spindle power, removes material from block **850** in two steps and following a toolpath which has an low stepover, typically 0.9 mm. This is shown at machining stages **880** and **882** and the toolpath is shown schematically and designated by reference numeral **884**.
[0306] It is appreciated that the foregoing description of FIG. **13** is merely illustrative of the functionality of an embodiment of the present invention and that additional parameters, such as feed speed and rpm may additionally be varied as a function of spindle power.
[0307] Reference is now made to FIG. **14**, which is a simplified pictorial illustration of tool paths for machining objects having thin walls in accordance with a preferred embodiment of the present invention.

[0308] The embodiment of the present invention described below with reference to FIG. **14** provides an automated computer-implemented system and method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object having a relatively thin wall from a workpiece. As illustrated, the system and method include automatically selecting, using a computer, a tool path having the following functional features: [0309] initially machining the workpiece at first maximum values of cutting depth, cutting width, cutting speed and cutting feed to have a relatively thick wall at the location of an intended relatively thin wall; [0310] reducing the height of the relatively thick wall to the intended height of the intended relatively thin wall; and thereafter [0311] reducing the thickness of the thick wall by machining the workpiece at second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values.

[0312] For the purposes of explanation, it is assumed that a workpiece, typically a block of steel **900**, having typical dimensions of 220 mm by 120 mm by 60 mm, is machined to produce a machined object **902** having internal pockets **904** each of dimensions 100 mm by 100 mm by 30 mm, separated by a thin wall **906** having a thickness of 1 mm and a height of 30 mm.

[0313] At a machining stage **910**, a toolpath **912** is followed, producing an initial cut **914** overlying the location of the thin wall **906** and having a width greater than the intended thickness of the thin wall **906**.

[0314] Thereafter, at a machining stage **920**, a toolpath **922** is followed producing two cuts **924** corresponding to internal pockets **904**.

[0315] Thereafter, at a machining stage **930**, a toolpath **932** is followed producing two cuts **934** defining an upper portion of thin wall **906**.

[0316] Thereafter, at a machining stage **940**, a toolpath **942** is followed producing two cuts **944**, below cuts **934**, defining a lower portion of thin wall **906**.

[0317] Finally, at a machining stage **950**, a toolpath **952** is followed reducing the height of the machined object **902** to the height of the thin wall **906**.

[0318] Reference is now made to FIG. **15**, which is a simplified, partially symbolic, partially pictorial illustration of exemplary selection of milling aggressiveness such as feed speed, rpm, axial depth of cut and stepover as a function of tool overhang in accordance with a preferred embodiment of the present invention. For the purpose of conciseness only stepover as a function of tool overhang is illustrated in FIG. **15**.

[0319] The embodiment of the present invention described below with reference to FIG. **15** provides an automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object. In the illustrated embodiment, the system and method include the following: [0320] ascertaining the extent of tool overhang of a tool being used in the computer numerically controlled milling machine; and [0321] automatically selecting, using a computer, a tool path which is a function of the tool overhang, the tool path having the following characteristics: [0322] for a first tool overhang selecting a tool path having first maximum values of cutting depth, cutting width, cutting speed and cutting feed; [0323] for a second tool overhang which is greater than the first tool overhang, selecting a tool path having second maximum values of cutting depth, cutting width, cutting speed and cutting feed, at least one of the second maximum values being less than a corresponding one of the first maximum values.

[0324] FIG. **15** shows a workpiece, typically a block of steel **954**, having typical dimensions of 300 mm by 300 mm by 150 mm, which is machined by each of two milling tools **955** and **956** to produce an identical machined object **957**. For clarity, the block **954** and the machined object **957** are shown out of proportion to the size of the milling tools **955** and **956**.

[0325] As seen in FIG. **15**, milling tool **955**, which has a relatively long tool overhang, typically 150 mm, removes material from block **954** following a toolpath which has a relatively small stepover, typically 1.5 mm. This is shown at machining stage **958** and the toolpath is shown

schematically and designated by reference numeral **959**.
[0326] As further seen in FIG. **15**, milling tool **956**, which has a relatively short tool overhang, typically 60 mm, removes material from block **954** following a toolpath which has a relatively large stepover, typically 3 mm. This is shown at machining stage **960** and the toolpath is shown schematically and designated by reference numeral **961**.
[0327] Reference is now made to FIG. **16**, which is a simplified pictorial illustration of machining functionality in accordance with another preferred embodiment of the present invention.
[0328] The embodiment of the present invention described below with reference to FIG. **16** provides an automated computer-implemented method and system for generating commands for controlling a computer numerically controlled milling machine to fabricate an object having a semi-open region. In the illustrated embodiment of the present invention, the method and system includes the following: [0329] estimating, using a computer, a first machining time for machining the semi-open region using a generally trichoidal type tool path; [0330] estimating, using a computer, a second machining time for machining the semi-open region using a generally spiral type tool path; and [0331] automatically selecting, using a computer, a tool path type having a shorter machining time.
[0332] For the purposes of explanation, it is assumed that a workpiece, typically a block of steel **962**, having typical dimensions of 150 mm by 120 mm by 50 mm, is machined to produce a machined object **963** having a semi-open pocket **964** of dimensions 140 mm by 100 mm by 20 mm
[0333] At a machining stage **970**, a toolpath **972** is followed, producing an initial cut **974** resulting in a closed pocket **976**. It is a particular feature of this embodiment of the present invention that the system initially machines the workpiece to define a geometry, here a closed pocket, which differs from the desired final geometry, here an semi-open pocket. This has an advantage in that it enables the machining of the pocket to be mainly done using a spiral tool path.
[0334] Thereafter, at a machining stage **980**, a toolpath **982** is followed, producing mutually spaced channels **984** extending through a wall **986**, which separates the closed pocket **976** from the edge of the workpiece. It is a particular feature of this embodiment of the present invention that the system initially machines channels in a wall which is ultimately to be removed.
[0335] Thereafter, at a machining stage **990**, a toolpath **992** is followed, removing blocks **994** which remained in the wall **986** following cutting of the channels **984** and thereby defining the semi-open pocket **964**.
[0336] Reference is now made to FIG. **17**, which is a simplified pictorial illustration of tool paths for machining objects having an hourglass-shaped channel in accordance with a preferred embodiment of the present invention.
[0337] The embodiment of the present invention described below with reference to FIG. **17** provides an automated computer-implemented system and method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object having a channel open at both its ends and including an intermediate narrowest portion. In the illustrated embodiment of the present invention the method includes automatically selecting, using a computer, a tool path type having first and second tool path portions, each starting at a different open end of the channel, the first and second tool path portions meeting at the intermediate narrowest portion.
[0338] For the purposes of explanation, it is assumed that a workpiece, typically a block of steel **1000**, having typical dimensions of 150 mm by 120 mm by 50 mm, is machined to produce a machined object **1002** having an hourglass-shaped channel **1004** which is open on both ends thereof.
[0339] At a machining stage **1006**, a toolpath **1008** is followed, producing an initial cut **1010**, resulting in an inwardly tapered semi open pocket **1012**.
[0340] Thereafter, at a machining stage **1016**, a toolpath **1018** is followed, producing a further cut **1020**, resulting in an inwardly tapered semi open pocket **1022** joined to pocket **1012** at a narrow

point and defining therewith hourglass-shaped channel **1004** which is open on both ends thereof.
[0341] It is a particular feature of this embodiment of the present invention that the system machines a channel in multiple stages so that thin wall machining takes place at a point of minimum width of the channel.
[0342] Reference is now made to FIG. objectobjectobject**18**, which is a simplified pictorial illustration of tool paths for machining objects which are calculated based on optimal cutting conditions for a maximum cutting depth and tool paths for cutting portions which involve cutting at less than the maximum cutting depth are calculated based on modified cutting conditions optimized for cutting at a depth less than the maximum cutting depth.
[0343] The embodiment of the present invention described below with reference to FIG. **18** provides an automated computer-implemented system and method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object which fabrication involves cutting a workpiece at least first and second different maximum depths of cut, wherein said first maximum depth of cut is greater than said second maximum depth of cut. In the illustrated embodiment of the present invention, the method includes automatically selecting, using a computer, at least first and second tool paths having corresponding first and second maximum values of cutting width, cutting speed and cutting feed, at least one of said second maximum values being greater than a corresponding one of said first maximum values.
[0344] For the purposes of explanation, it is assumed that a workpiece, typically a block of steel **1200**, having typical dimensions of 150 mm by 120 mm by 50 mm, is machined to produce a machined object **1202** having a semi-open pocket **1204** having a ledge **1206**.
[0345] At a machining stage **1210**, a toolpath **1212** is followed, producing an initial cut **1214**, resulting in a semi-open pocket **1216**. Toolpath **1212** has a relatively small stepover, typically 1.5 mm.
[0346] Thereafter, at a machining stage **1220**, a toolpath **1222** is followed, producing a cut **1224**, resulting in a semi-open ledge **1226**. Toolpath **1222** has a relatively large stepover, typically 3 mm.
[0347] It is a particular feature of this embodiment of the present invention that the system adjusts the cutting conditions to ensure that the mechanical load experienced by the milling tool is at a generally constant optimized value.
[0348] Reference is now made to FIGS. **19**A, **19**B and **19**C, which are simplified illustrations of three alternative tool repositioning moves from the end of one tool path to the beginning of a subsequent tool path in the same pocket in accordance with a preferred embodiment of the present invention. Each of FIGS. **19**A, **19**B and **19**C includes a pictorial view and a top view, for the purpose of clarity.
[0349] The embodiment of the present invention described below with reference to FIGS. **19**A-**19**C provides an automated computer-implemented system and method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object, wherein fabrication of the object involves calculating multiple tool paths requiring tool repositioning therebetween along a selectable repositioning path. In the illustrated embodiment of the present invention, the method includes the following steps which can take place multiple times, each for a different repositioning between tool path segments: [0350] estimating, using a computer, a first repositioning time for a first repositioning path which includes travel in a clearance plane above a workpiece; [0351] estimating, using a computer, a second repositioning time for a second repositioning path which does not include tool travel in the clearance plane; and [0352] automatically selecting, using a computer, a repositioning path having a shortest repositioning time.
[0353] For the purposes of explanation, there is illustrated a partially machined workpiece **1300**, having typical maximum outer dimensions of 150 mm by 120 mm by 50 mm. As seen, the workpiece **1300** has formed therein a closed pocket **1302** having an island **1304** formed with a slot **1306** extending therethrough. Machining of a first corner ledge **1310**, involving a first tool path segment, has been completed and the tool must be repositioned to machine a second corner ledge

1312, involving a second tool path segment.
[0354] In FIG. **19**A the tool follows a repositioning path **1320** which draws the tool initially upwardly from ledge **1310** beyond the top of the workpiece **1300** and then across the workpiece at a uniform height in a straight line to a location above ledge **1312** to be machined and then downwardly to machine ledge **1312**. The tool travel time for repositioning path **1320** is calculated.
[0355] In FIG. **19**B the tool follows an alternative repositioning path **1330** which draws the tool without changing its height, from ledge **1310** around island **1304** to the appropriate location of the tool for milling ledge **1312**. The tool travel time for repositioning path **1330** is calculated.
[0356] In FIG. **19**C the tool follows a further repositioning path **1340** which draws the tool initially upwardly from ledge **1310** to a minimum height, below the top of the workpiece **1300**, at which height the tool clears the island **1304** and then across the workpiece at this height, not necessarily in a straight line to a location above ledge **1312** to be machined and then downwardly to machine ledge **1312**. The tool travel time for repositioning path **1340** is calculated. In this case, the repositioning path **1340** passes through slot **1306** and is not necessarily collinear with a line connecting the locations of ledges **1310** and **1312**. The tool travel time for repositioning path **1340** is calculated.
[0357] The tool travel times for the above three reposition paths are compared and the repositioning path having the shortest tool travel time is employed.
[0358] It is a particular feature of this embodiment of the present invention that the repositioning path having a minimum tool travel time is employed.
[0359] It will be appreciated by persons skilled in the art that the present invention is not limited by what has been particularly claimed hereinbelow. Rather the scope of the present invention includes various combinations and subcombinations of the features described hereinabove as well as modifications and variations thereof as would occur to persons skilled in the art upon reading the foregoing description with reference to the drawings and which are not in the prior art.

Claims

1-37. (canceled)
38. An automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object, wherein fabrication of the object involves calculating multiple tool paths requiring tool repositioning therebetween along a selectable repositioning path, the method comprising: estimating, using a computer, a first repositioning time for a first repositioning path which includes travel in a clearance plane above a workpiece; estimating, using a computer, a second repositioning time for a second repositioning path which does not include tool travel in said clearance plane; and automatically selecting, using a computer, a repositioning path having a shortest repositioning time.
39. An automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object according to claim 38 and wherein said second repositioning path is automatically selected by said computer from among possible multiple repositioning paths which do not include tool travel in said clearance plane on the basis of shortest repositioning time.
40. An automated computer-implemented method for generating commands for controlling a computer numerically controlled milling machine to fabricate an object according to claim 39 and wherein said multiple repositioning paths include repositioning paths which require raising of said tool and repositioning paths which do not require raising of said tool.
41-73. (canceled)
74. An automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate an object, wherein fabrication of the object involves calculating multiple tool paths requiring tool repositioning therebetween along a

selectable repositioning path, the apparatus comprising a tool path configuration engine operative for: estimating, using a computer, a first repositioning time for a first repositioning path which includes travel in a clearance plane above a workpiece; estimating, using a computer, a second repositioning time for a second repositioning path which does not include tool travel in said clearance plane; and automatically selecting, using a computer, a repositioning path having a shortest repositioning time.

75. An automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate an object according to claim 74 and wherein said second repositioning path is automatically selected by said computer from among possible multiple repositioning paths which do not include tool travel in said clearance plane on the basis of shortest repositioning time.

76. An automated computer-implemented apparatus for generating commands for controlling a computer numerically controlled milling machine to fabricate an object according to claim 75 and wherein said multiple repositioning paths include repositioning paths which require raising of said tool and repositioning paths which do not require raising of said tool.